space structures laboratory

Scalability of Prestressed Structures Assembled in Space

Tara Venkatadri

PhD Candidacy Presentation

February 5th, 2026

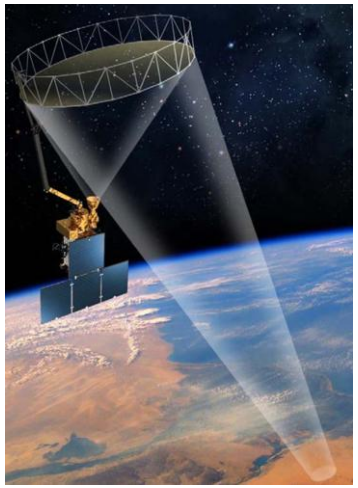
Committee: Professor Sergio Pellegrino (Advisor)
Professor Guruswami Ravichandran (Chair)
Professor Daniel Meiron
Professor Kaushik Bhattacharya

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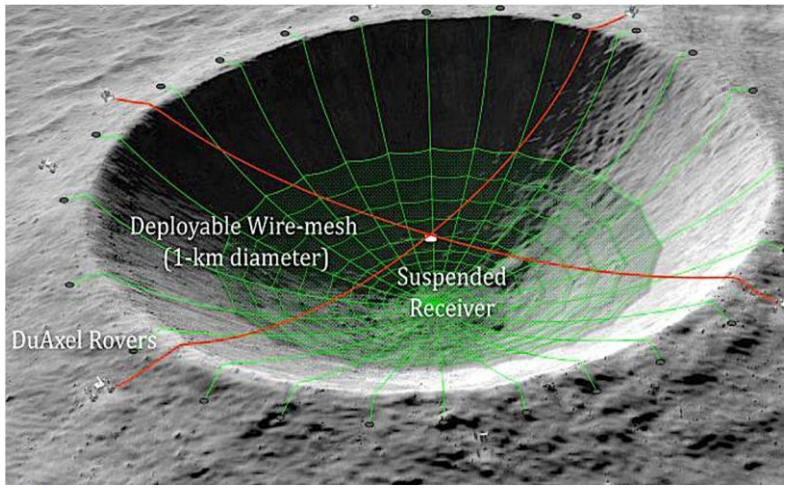
Large Reflectors in Space

- Remote sensing applications require space-based reflector antennas
- Benefits of increased size: Improved resolution, sensing of fainter objects

SMAP Radar: 6 m [1]

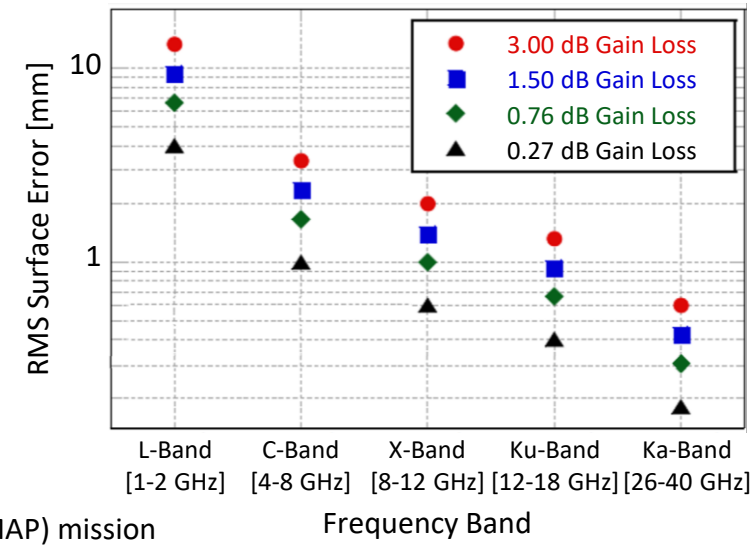


Lunar Crater Radio Telescope: 1 km [2]

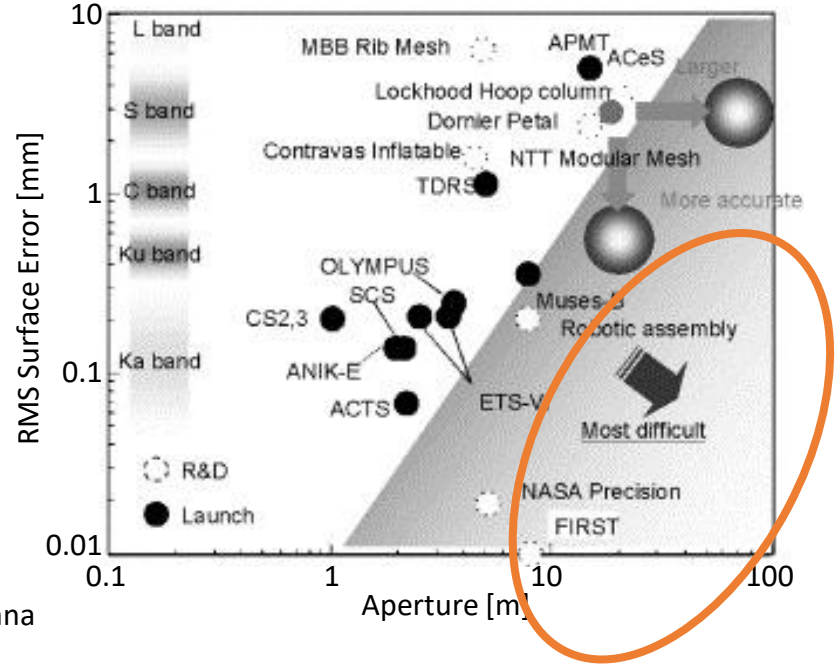


- Paraboloid surface precision impacts performance
- Shape accuracy and transport to space are challenging to achieve as size increases

Surface Precision Performance Effects [3]



Past Missions: Diameter vs. Accuracy [4]



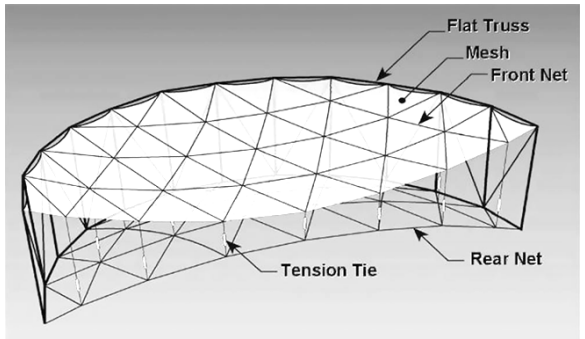
[1] Entekhabi, D. et al. (2010). The Soil Moisture Active Passive (SMAP) mission
[2] Bandyopadhyay, S., et al. (2021). Conceptual design of the Lunar Crater Radio Telescope (LCRT) on the far side of the Moon
[3] Park, T. Y., et al. (2021). Thermal design and analysis of unfurlable CFRP skin-based parabolic reflector for spaceborne SAR antenna
[4] Meguro, A., et al. (2003). Key technologies for high-accuracy large mesh antenna reflectors

Scaling of Prestress and Assembly Methods

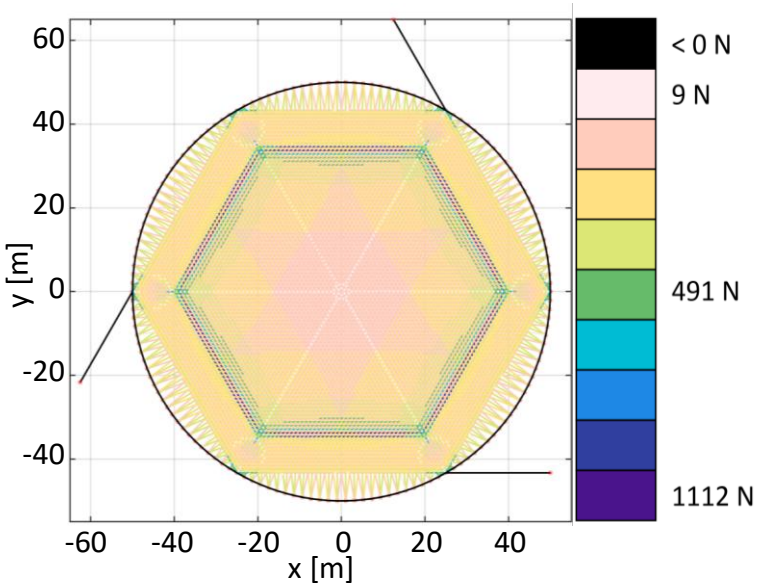
Shape Accuracy

- Prestressed (fully tensioned) cable net with perimeter truss
- Triangular facet size impacts precision
- Larger diameters require more cable rings and perimeter polygon sides

AstroMesh Reflector [5]



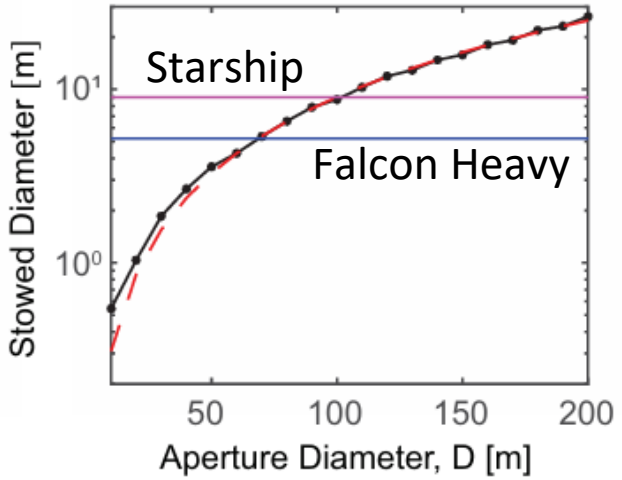
Stress Concentrations



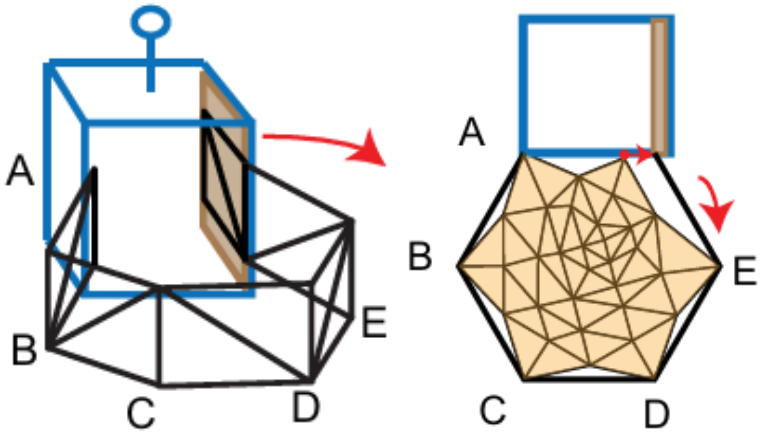
Transport to Space

- Large deployable reflectors exceed launch vehicle capacity
- Solution: Robotic in-space assembly
- Stepwise and final shapes depend on kinematic constraints, increased sensitivity for many-sided structures

Stowed Deployable Antenna Scaling [6]



Stationary Robot [7]



[5] Shi, H., et al. (2018). New methodology of surface mesh geometry design for deployable mesh reflectors
[6] Suh, J. E., et al. (2024). Scaling laws for deployable mesh reflector antennas
[7] Suh, J. E., et al. (2024). Concept for scalable mesh reflector antennas assembled in space

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Problem Statement and Research Objectives

Problem Statement

Extend the design and assembly of ring-like space structures beyond current scaling limits

Research Objectives

Problem 1

Prestressability of final cable net

- Examine effect of stress concentrations
- Increase number of rings in tessellation

Problem 2

Feasibility of stationary robot assembly

- Examine effect of kinematic constraints
- Increase number of sides in perimeter

Methodology to Date

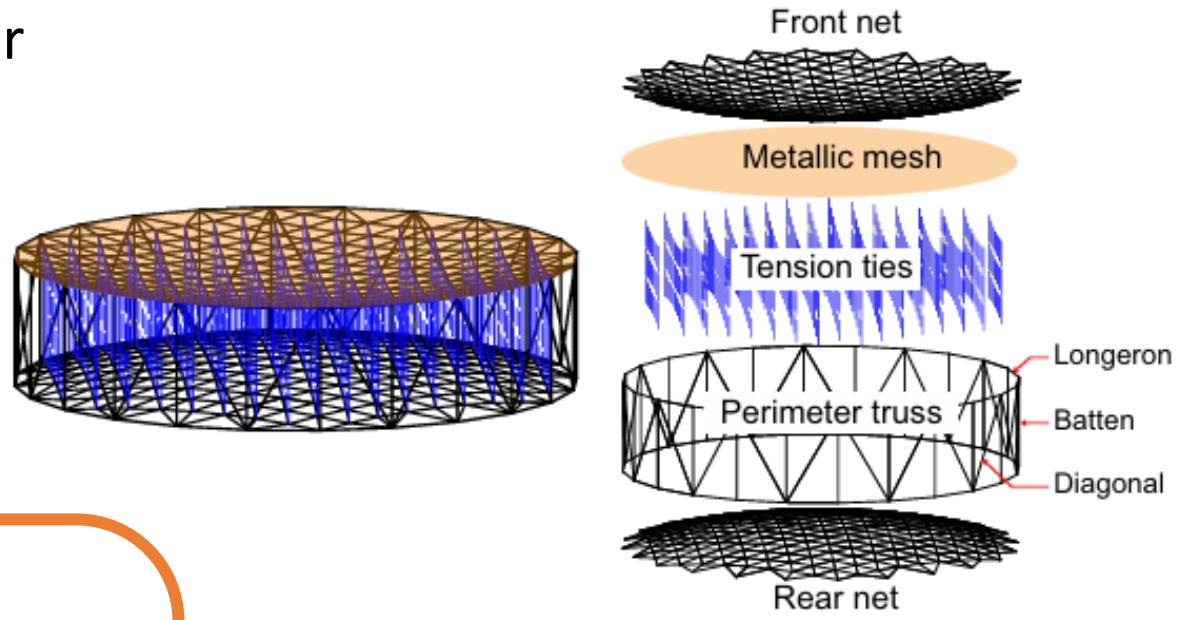
- Cable topology for strut-cable structure
- MATLAB optimization-based form and force finding [8]
- Enhancement of prestress and assembly method scalability

[8] Adriaenssens, S., et al. (2014). Shell structures for architecture: Form finding and optimization

Cable Net Tessellation and Prestressing

- Goal: Determine limiting number of rings (N) for prestressability
- Cable net design:
  - Inner net: Hexagonal equilateral triangle tessellation, paraboloid projection
  - Boundary net: Connects battens to inner net

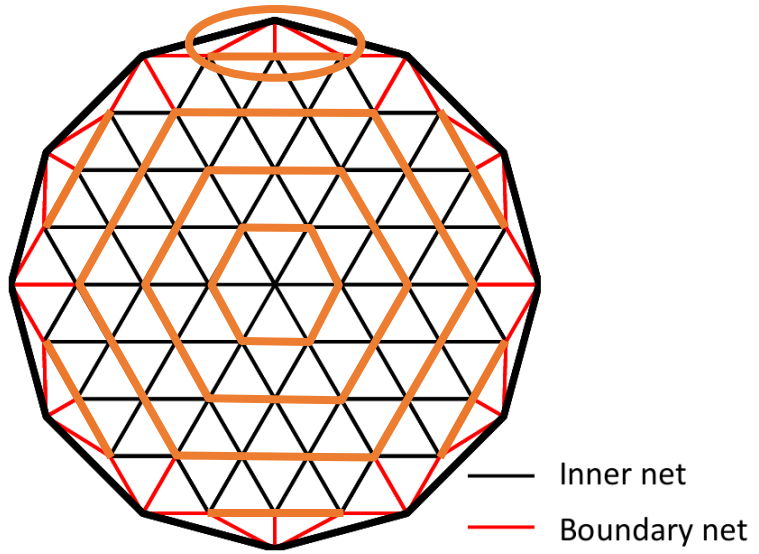
Components of Antenna [6]



Prestress Optimization Algorithm [6]

- Tension tie forces x grouped by ring
- Member force basis $B = [B_c\ B_l\ B_b]^T$
- Objective: Minimize batten force, $\frac{-1}{n_b}\ [1, 1, \dots, 1] B_b x$
- Constraints: $B_c x \geq c_{min}$
- MATLAB linprog simplex algorithm [9]

4-Ring Cable Net [6]

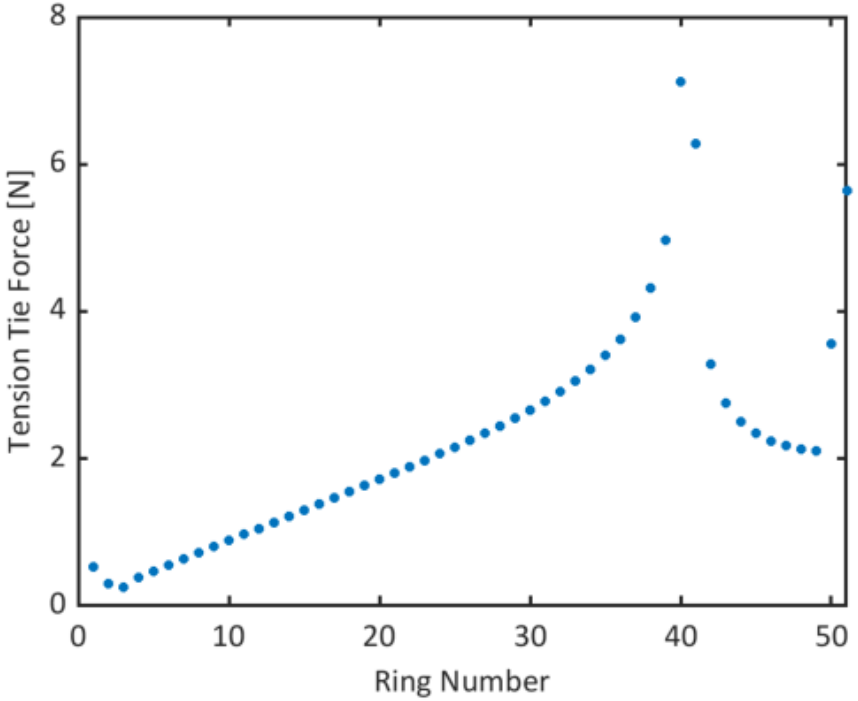


[6] Suh, J. E., et al. (2024). Scaling laws for deployable mesh reflector antennas
[9] Nocedal, J., & Wright, S. J. (2006). Numerical optimization

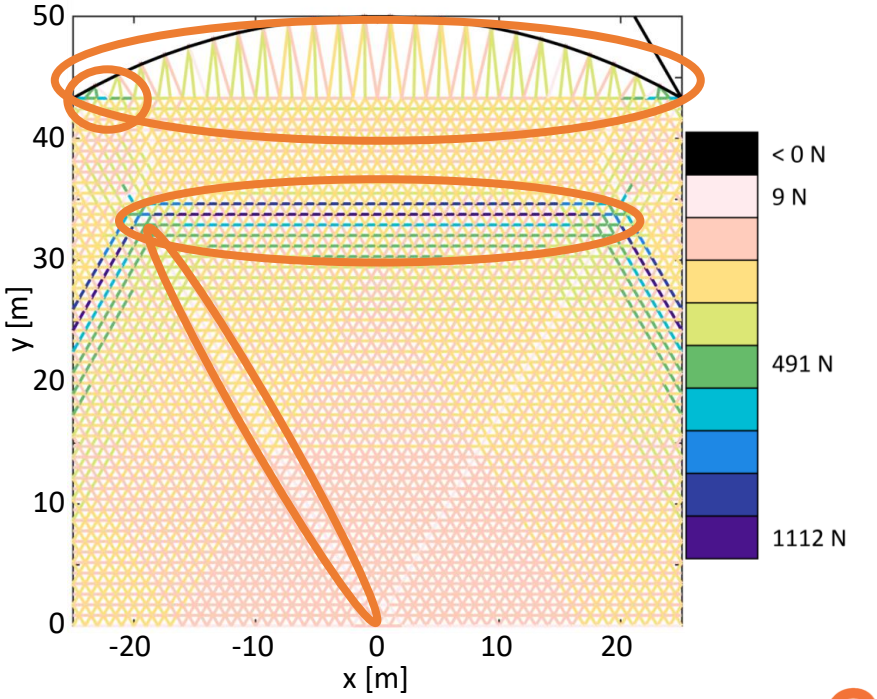
Limiting Number of Rings (N) for Existing Method

- Linear programming infeasible above $N = 50$ rings
- Stress concentrations on interior peak ring, hexagon-circle transition, hexagon vertices, and boundary cables
  - Same cables become compressive for $N > 50$
- Need to shorten boundary cables and smooth hexagon-circle transition

Tension Tie Forces: $N = 50$



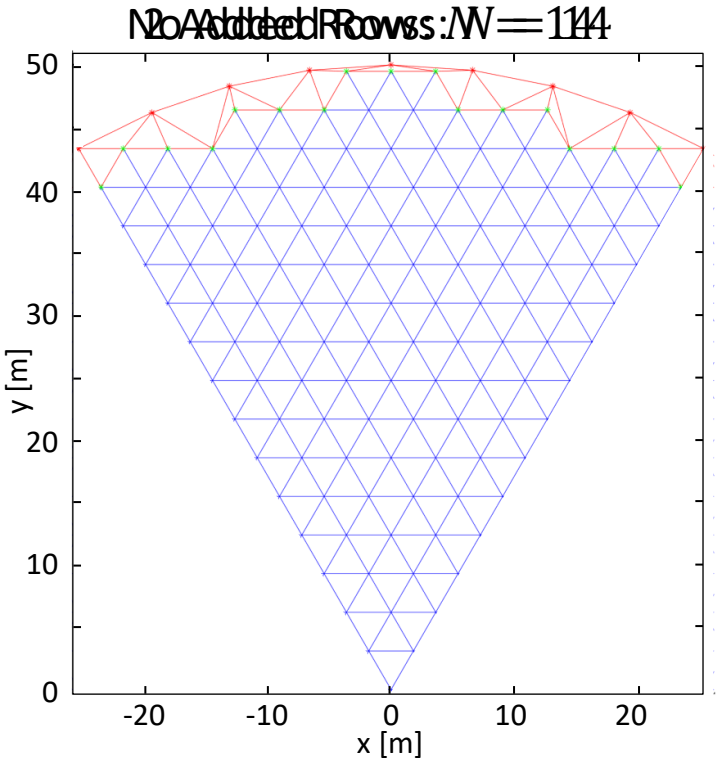
Cable Forces: $N = 50$



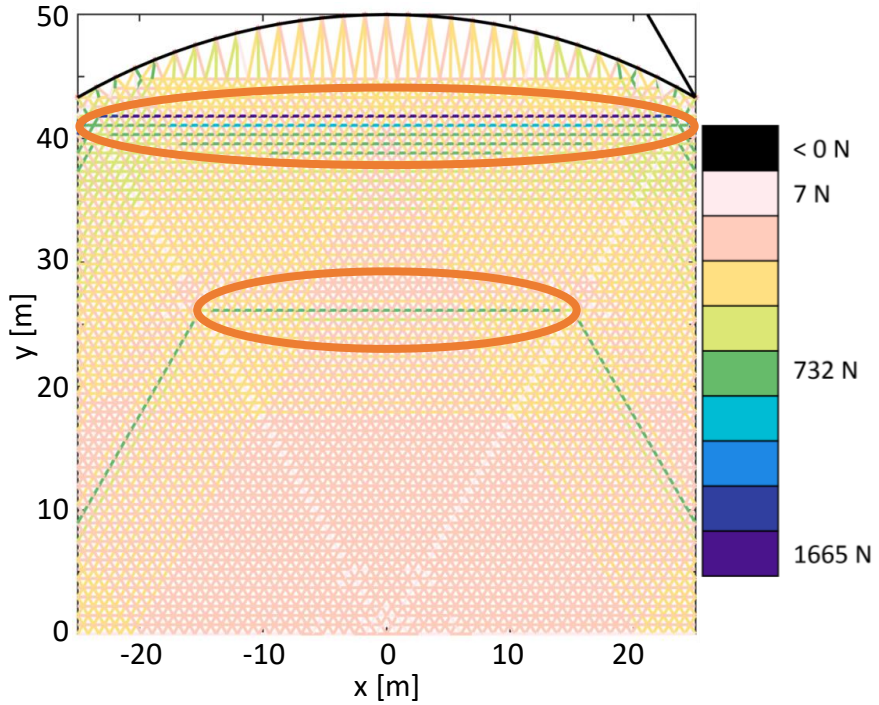
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Tessellation Adjustment

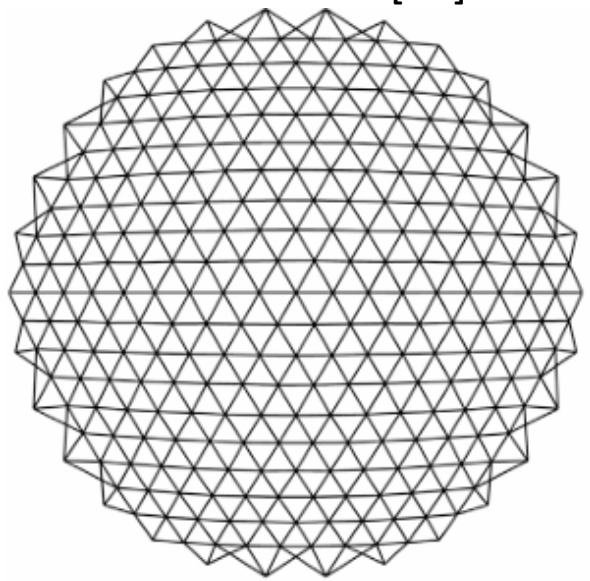
- Adjustment: Fill in boundary region with partial rows of triangles
- Adding 2 rows enables convergence up to $N = 58$ rings
- Reaching limits of 2D hexagonal tessellation projected into 3D
  - Issues: Sharp corners creating stress concentrations and distortion from projection
- Future work: Consider curved-surface tessellations that address AstroMesh limitations



Cable Forces: $N = 58$



Geodesic Tessellation on a Paraboloid [10]



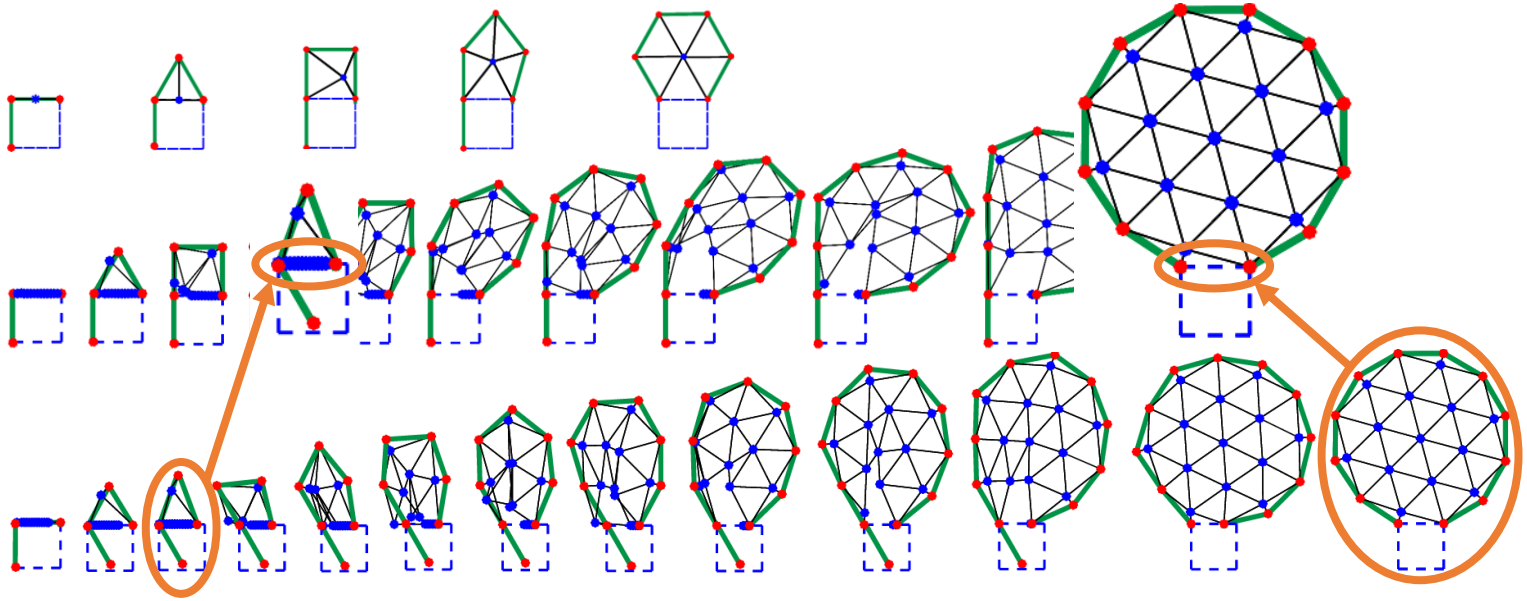
[10] Deng, H., et al. (2016). Design of geodesic cable net for space deployable mesh reflectors

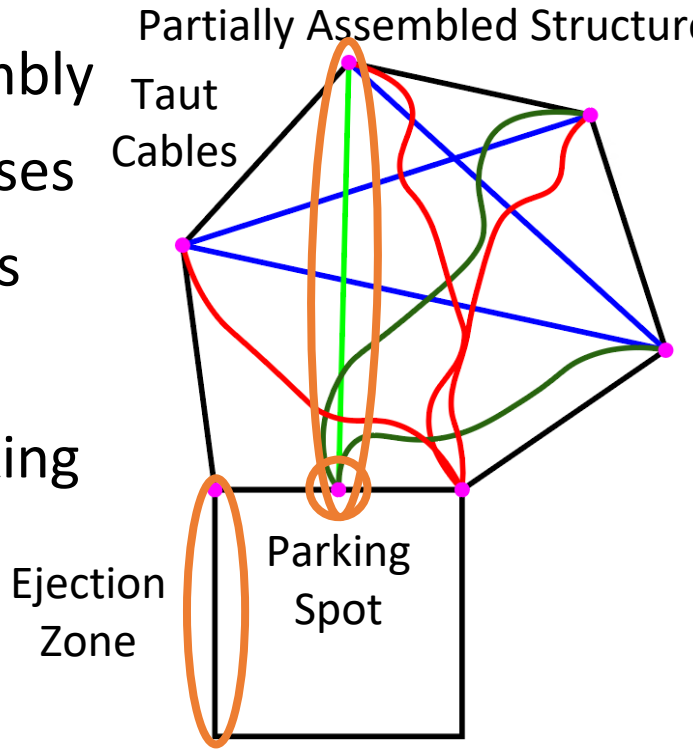
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Stationary Robot Assembly Concept

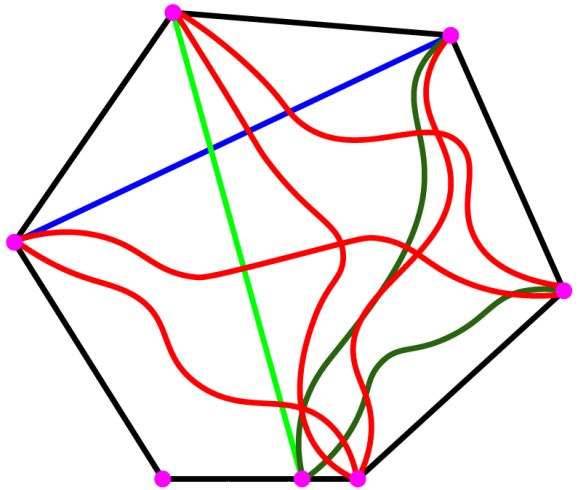
- Goal: Determine limiting number of sides (s) for feasible assembly
- Adjustments to assembly constraints are necessary as s increases
- Need simplified kinematic model to study various cable designs
- Initial study: Modified cable topology (cables on diagonals)
- Key features: Discrete strut ejection, cable parking spot, shrinking top edge of working area to reduce number of taut cables

Finite Element Model of AstroMesh Assembly (6 and 12 Sides) [11]





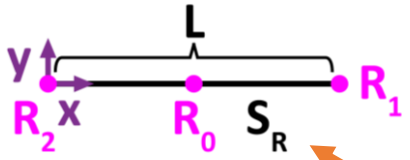
Shortened Edge and Slack Cables



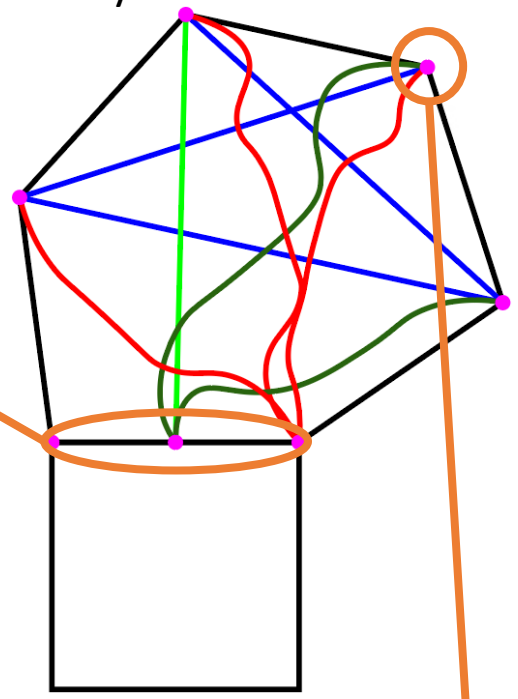
[11] Dassanayake, S. P., et al. (2025). Numerical study of novel concept for in-space assembly of ring-like space structures

Numerical Assembly Model Setup

Stationary Robot Notation



Partially Assembled Structure

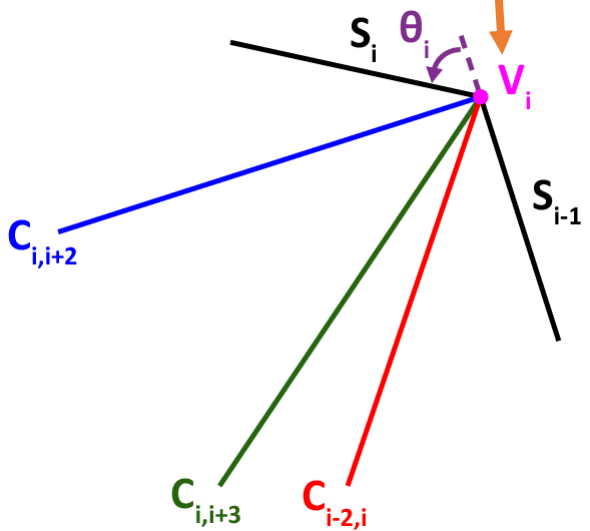


- R_0 and R_1 are parameters in model
- $(R_0, R_1) = (0, L)$ provides feasible regular polygon sequence for any s
  - $R_0 = 0$ is impractical for physical components of finite size
  - Need to study $R_0 > 0$, requires numerical model

Strain Energy Optimization Algorithm

- Objective: Minimize torsional spring energy, $\frac{1}{2}\alpha \sum_{i=1}^{s} \theta_i^2$
  - $\theta_i = \arccos\left(\frac{\vec{S}_i \cdot \vec{S}_{i-1}}{|\vec{S}_i||\vec{S}_{i-1}|}\right)$ and $\vec{S}_i = \begin{pmatrix} V_{i+1,x} - V_{i,x} \\ V_{i+1,y} - V_{i,y} \end{pmatrix}$
- Constraints: $\left\|\vec{S}_i\right\| = L$, $\left\|\vec{C}_{r,t}\right\| \leq L_C$
- MATLAB fmincon sequential quadratic programming algorithm [9]

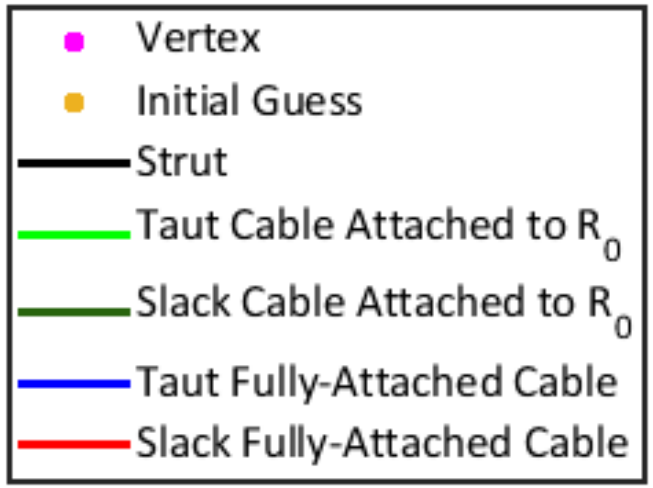
Vertex Notation



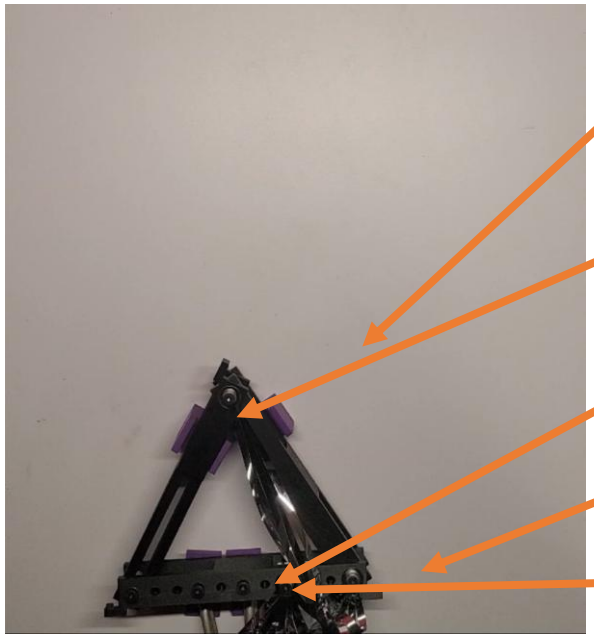
[9] Nocedal, J., & Wright, S. J. (2006). Numerical optimization

Octagon Assembly Sequence

- Focus on $R_1 = L$ to assess effect of R_0
- Initial results: Concave penultimate and final shape for any $R_0 > 0$
- Applied convexity constraint (angle stops)
- Optimization only feasible for $R_0 \geq 0.623L$
- Demonstrates effect of kinematic constraints on final shape as sides are added



PLA/Mylar Prototype:
$R_0 = 0.7L$, Angle Stops



Regular polygon due to slack cables

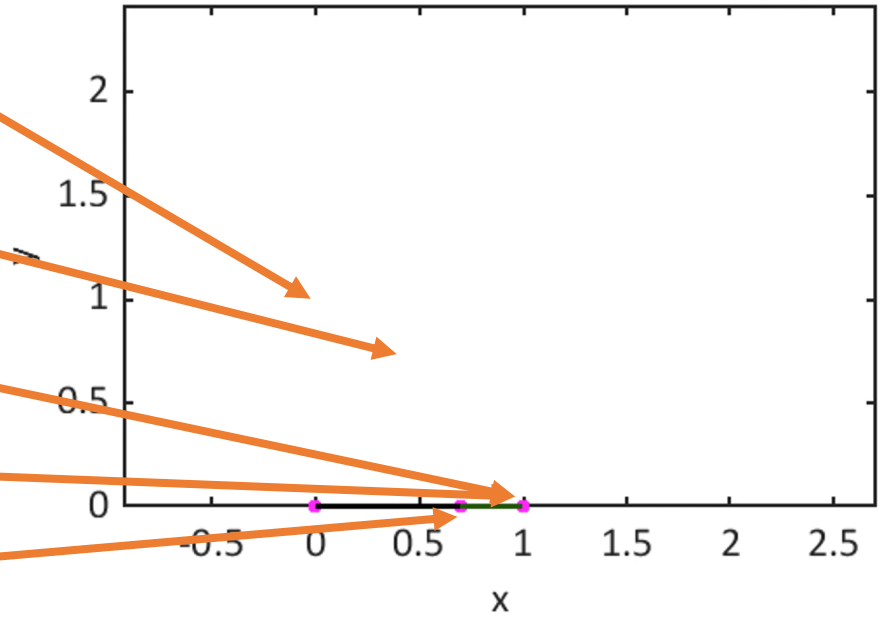
Irregular shape due to taut cables

Angle stops active

Desired final shape

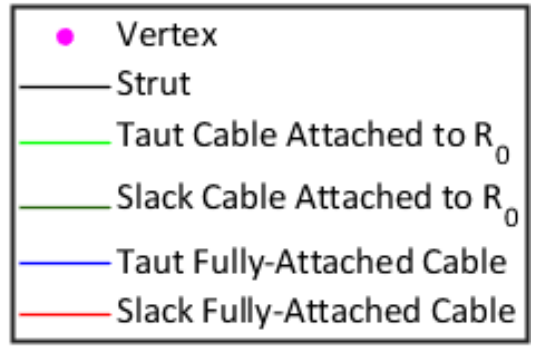
R_0

MATLAB Model:
$R_0 = 0.7L$, Convexity Constraint

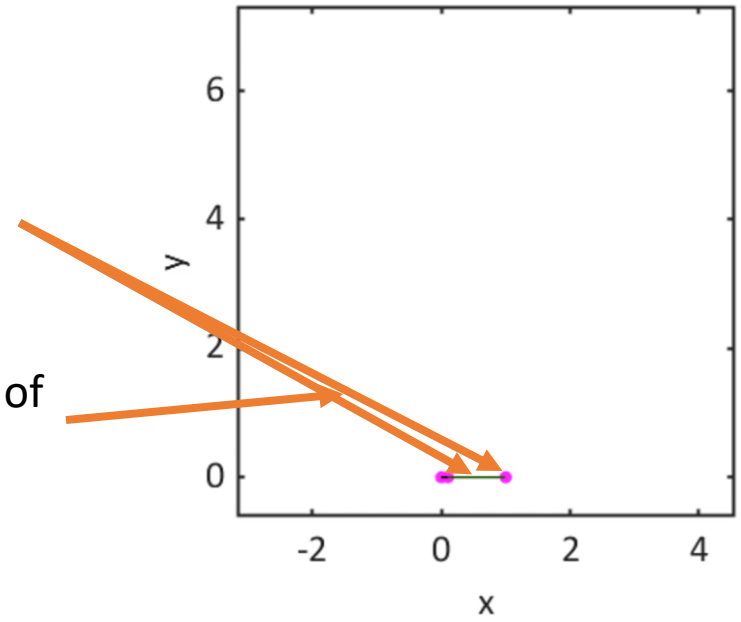


Limiting Number of Sides (s)

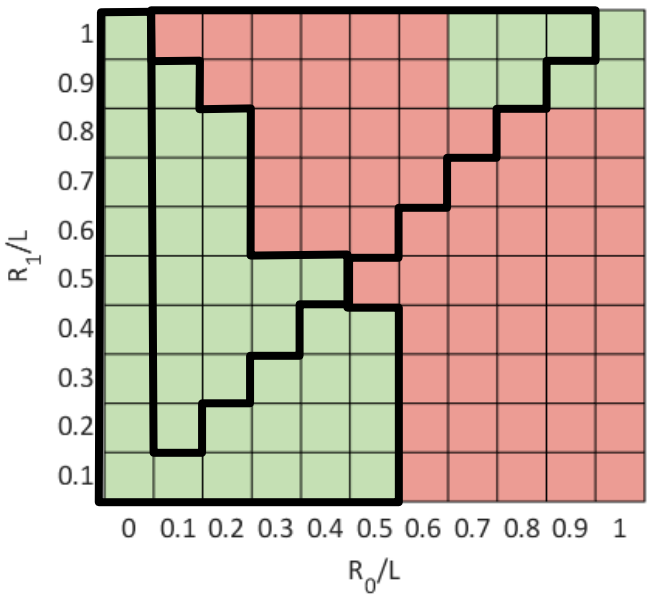
- $R_1 = L$: Infeasible optimization for any $R_0 > 0$ for $s \geq 9$, need to assess effect of moving R_1
- Performed (R_0, R_1) grid searches to establish feasibility
  - Cases with $0 < R_0 < R_1$ are most practical for implementation
  - Range of valid solutions (and maximum R_0) shrinks with staircase structure as s increases
  - $s = 23$: Maximum for which $R_0 = 0.1L$ provides valid solutions
- Future work: Increase physical realism, consider cable nets



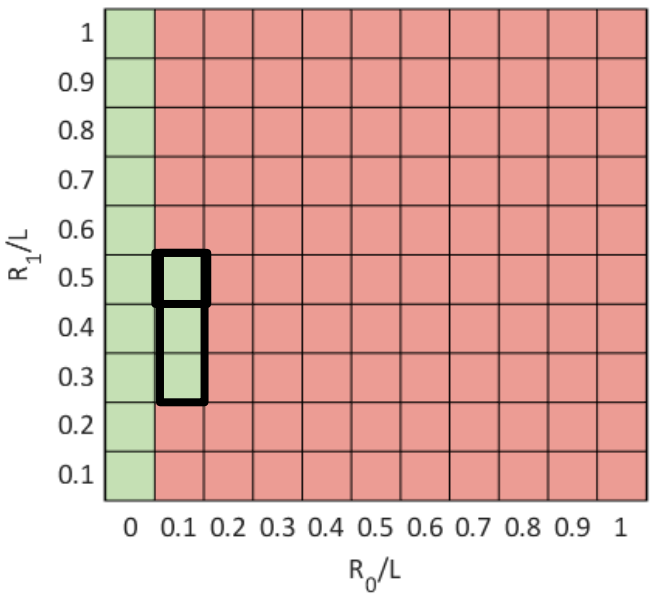
Assembly Sequence: $s = 23$, $R_0 = 0.1L$, $R_1 = 0.5L$



$s = 8$ Grid Search



$s = 23$ Grid Search



Movement of R_1

Delayed formation of taut cables

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Conclusion and Future Work

Summary:

- Established limiting number of rings for a simple prestressed cable tessellation
  - Identified sources of stress concentrations to avoid in future
- Demonstrated application of stationary robot method to many-sided structures
  - Modified cable topology to study adjustments/constraints for successful assembly

Future Work:

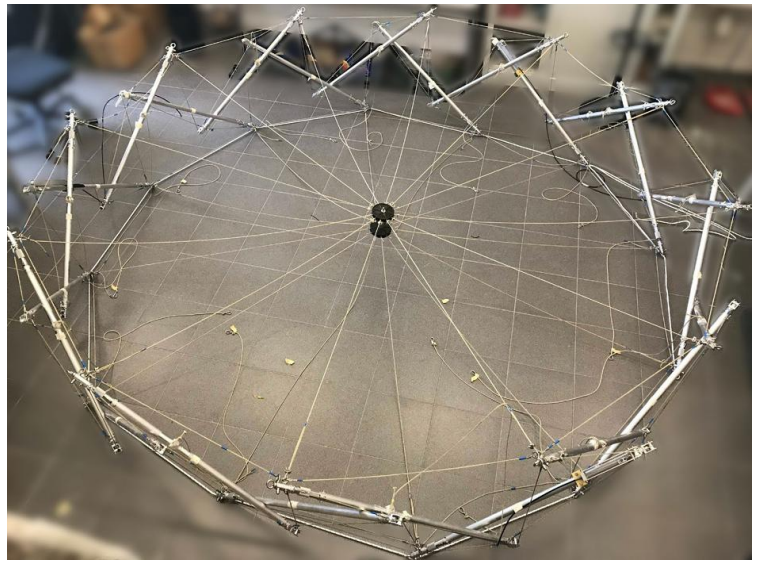
Model Scalability

- Improve cable net topology and assembly model realism
- Test prestressability and assembly feasibility of giant cable net reflectors

Alternative Architectures

- Prestress non-paraboloid cable nets (surfaces for other space structures)
- Assemble non-prismatic 3D structures (e.g. tensegrity rings)

Tensegrity Ring [12]



Acknowledgements: Dr. Jong-Eun Suh, Dr. Sahangi Dassanayake, DARPA NOM4D, NSF GRFP

[12] Gómez-Jauregui, V., et al. (2023). Tensegrity applications to architecture, engineering and robotics: A review

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1. Entekhabi, D., Njoku, E. G., O'Neill, P. E., Kellogg, K. H., Crow, W. T., Edelstein, W. N., ... & Van Zyl, J. (2010). The Soil Moisture Active Passive (SMAP) mission. *Proceedings of the IEEE*, *98*(5), 704-716.
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3. Park, T. Y., Kim, S. Y., Yi, D. W., Jung, H. Y., Lee, J. E., Yun, J. H., & Oh, H. U. (2021). Thermal design and analysis of unfurlable CFRP skin-based parabolic reflector for spaceborne SAR antenna. *International Journal of Aeronautical and Space Sciences*, *22*(2), 433-444.
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12. Gómez-Jauregui, V., Carrillo-Rodríguez, Á., Manchado, C., & Lastra-González, P. (2023). Tensegrity applications to architecture, engineering and robotics: A review. *Applied Sciences*, *13*(15), 8669.

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Thank you!
Questions?

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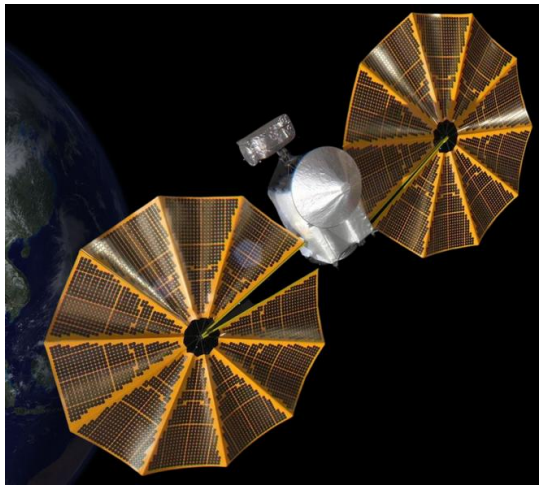
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Backup Slides

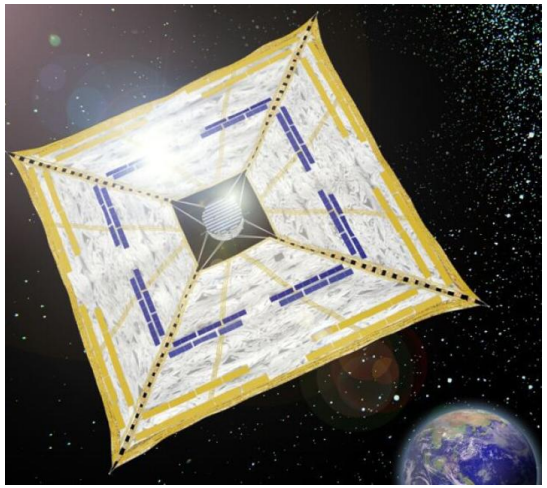
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Introduction: Other Ring-Like Space Structures

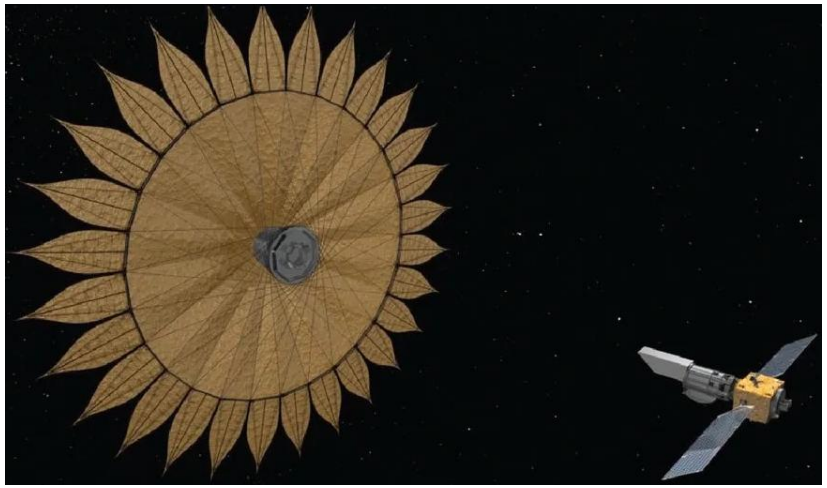
Lucy Solar Array [1]



IKAROS Solar Sail [2]



Starshade [3]

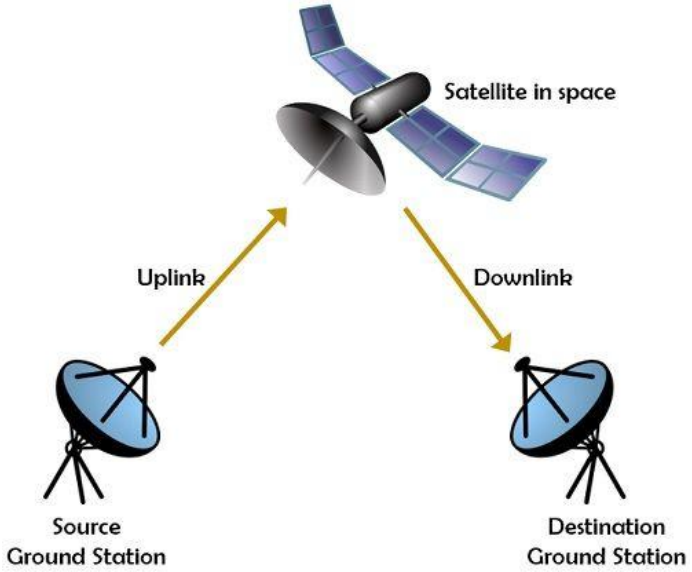


- Advantages of increased size: Larger area enabling more power generation and increased acceleration, better starlight occultation improving exoplanet imaging
- Operational shape requirements: Flat solar panels to increase projected area facing Sun, flat solar sails for efficient propulsion in a given direction, starshade petals to reduce diffraction of starlight around edges

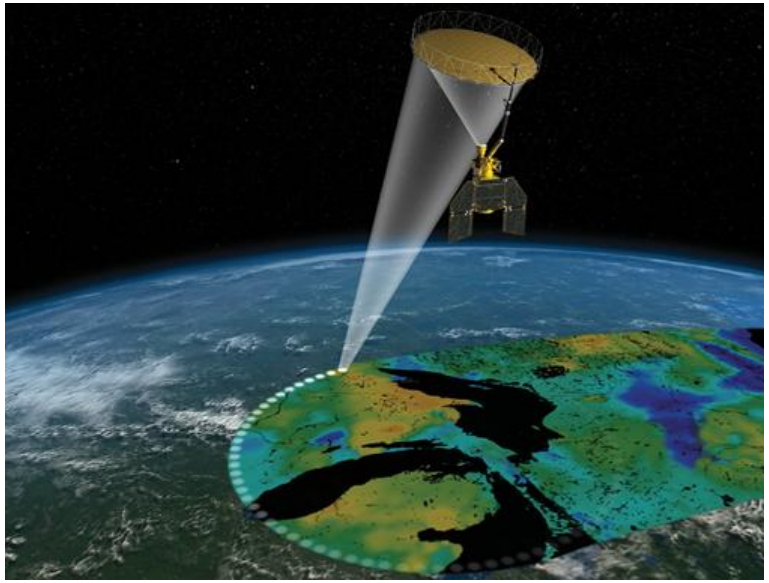
[1] Lounsbury, J., & Sekerak, M. (2025). Post-launch verification of Lucy solar array deployment
[2] Okuizumi, N., et al. (2021). Analytical investigation of global deployed shape of a spinning solar sail membrane
[3] Arya, M., et al. (2021). Demonstration of deployment repeatability of key subsystems of a furled starshade architecture

Introduction: Applications of Reflector Antennas in Space

Satellite Communication with Ground Stations



Remote Sensing Radar and Radiometry Applications [4]



Space-Based Radio Telescopes

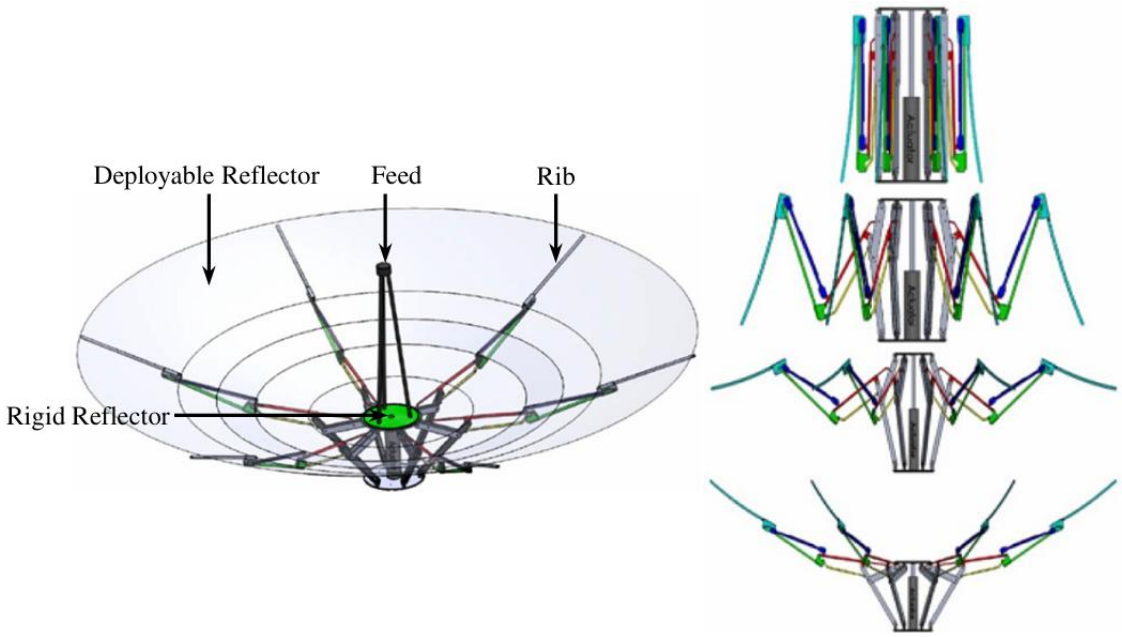


- Advantages of increasing antenna diameter:
  - Stronger communication signals for transmission over longer distances
  - Ability to collect more precise data and detect fainter signals for planetary science and astrophysical remote sensing applications

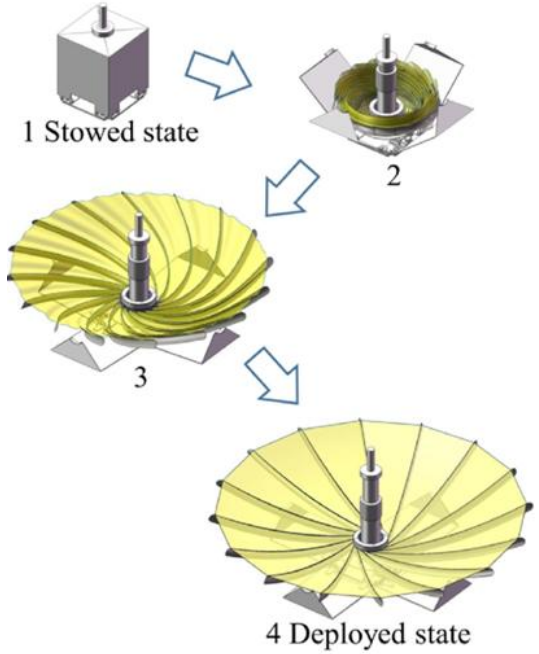
[4] Kellogg, K., et al. (2013). NASA's Soil Moisture Active Passive (SMAP) observatory

Introduction: Alternative Reflector Architectures

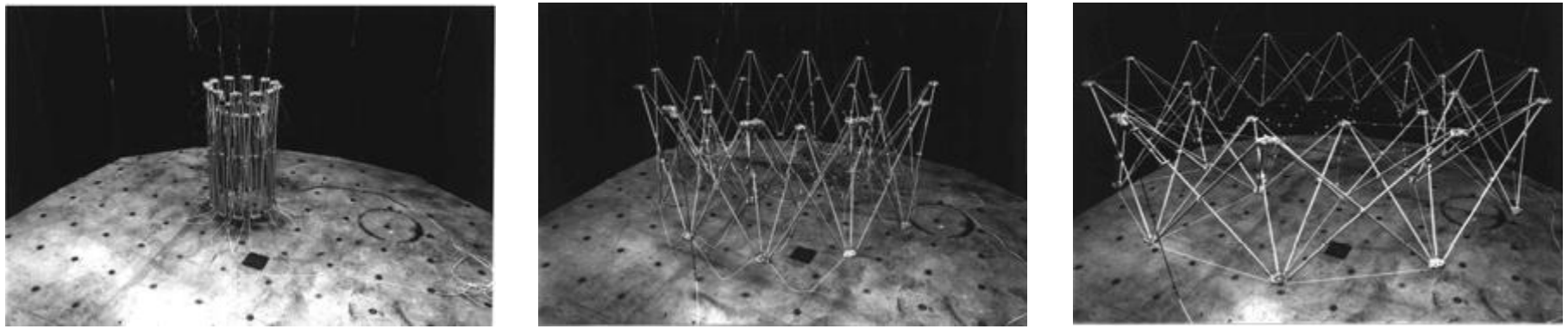
Umbrella [5]



Wrap-Rib [6]



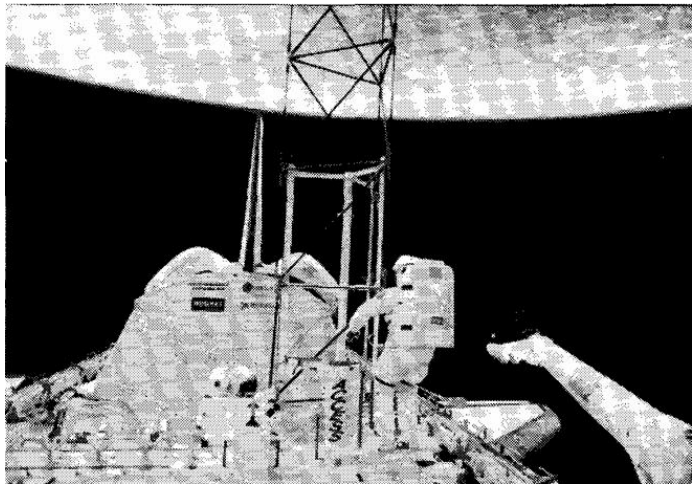
Pantograph [7]



[5] Hasanzade, V., et al. (2017). Compact deployable umbrella antenna design with optimum communication properties
[6] Zhang, H., et al. (2025). Design, analysis, and test of an ultra-compact X-band deployable wrap-rib antenna
[7] You, Z., & Pellegrino, S. (1997). Cable-stiffened pantographic deployable structures part 2: Mesh reflector

Introduction: Human-Operated In-Space Assembly Schemes

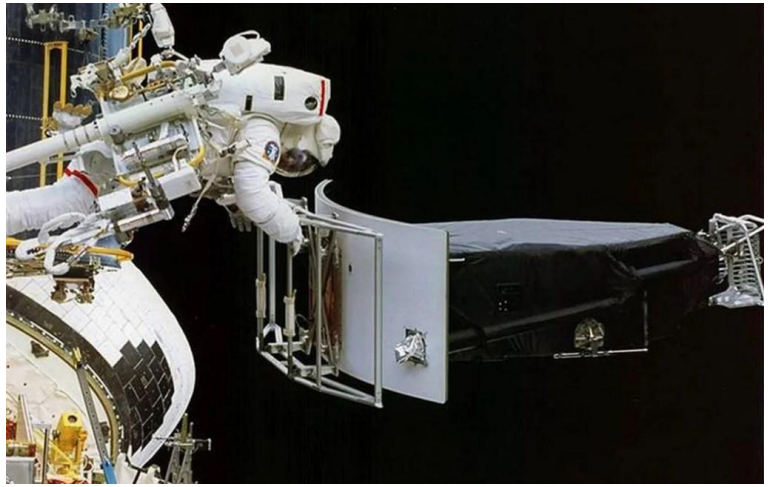
ACCESS: Truss Construction in Space Shuttle Payload Bay [8]



International Space Station Construction



Hubble Telescope Servicing

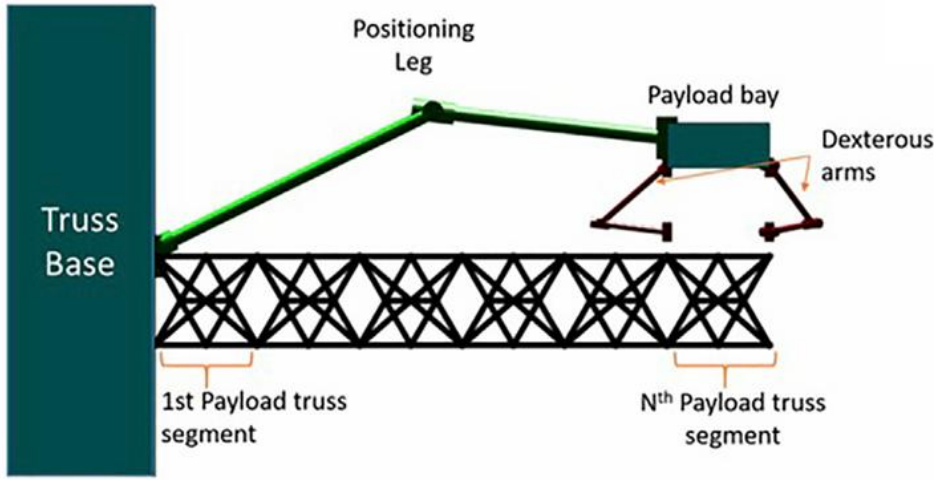


- Limitations of astronaut ISA [9]:
  - Neutral buoyancy training time
  - EVA time limits and astronaut fatigue
  - Astronaut force and dexterity limits
  - Cost/risk of life support systems
- Robots in orbit teleoperated from ground stations have also been considered [10]
  - Issues with latency in communication, will be exacerbated with increased distance from Earth

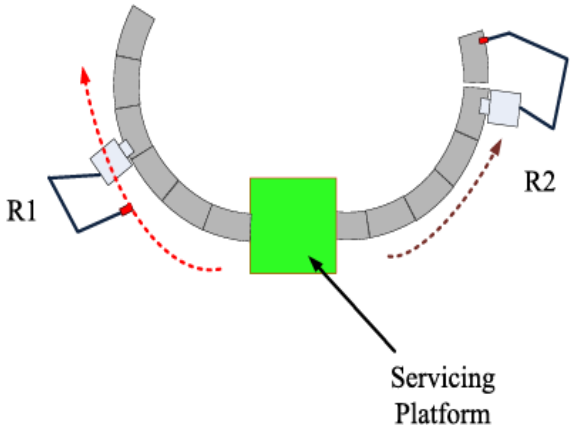
[8] Heard, Jr, W., et al. (1986). Results of the ACCESS space construction Shuttle flight experiment
[9] Heard Jr, W. L., & Lake, M. S. (1994). Neutral buoyancy evaluation of extravehicular activity assembly of a large precision reflector
[10] Ueno, H., et al. (1997). On-orbit construction experiment by tele-operated robot arm

Introduction: Robotic In-Space Assembly Schemes

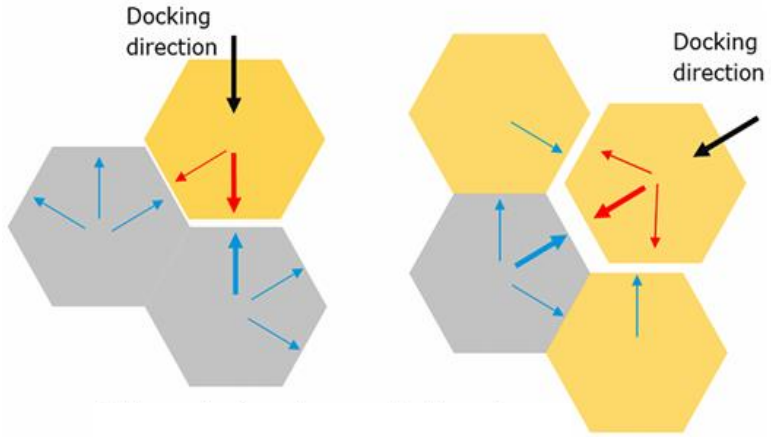
Long-Reach Stationary Robot Arm [11]



Assembly Robot Traversing Structure [12]



CubeSat Rendezvous and Docking [13]

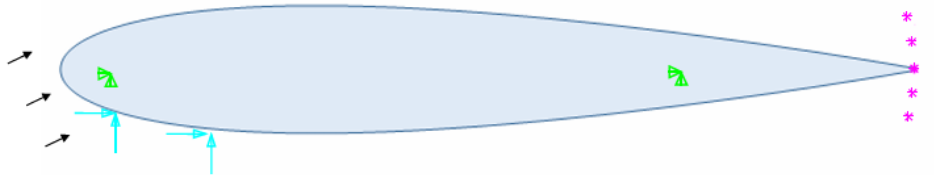


- Potential modifications to basic assembly methods:
  - Manufacturing structure on turntable to allow greater reach for stationary robot [14]
  - Single robot vs. multiple robots working in tandem (requires path planning) [15]
  - Formation flying to increase effective size of multi-part structures [16]

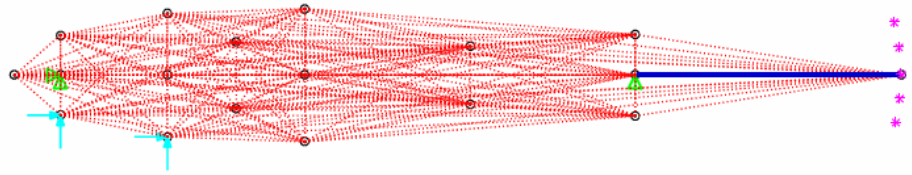
[11] McBryan, K. (2022). Comparing multi-arm robotics for in-space assembly
[12] Cao, K., et al. (2021). Dynamics and on-orbit assembly strategies for an orb-shaped solar array
[13] Pirat, C., et al. (2022). Toward the autonomous assembly of large telescopes using CubeSat rendezvous and docking
[14] Chapin, S., et al. (2023). Built On-orbit Robotically assembled Gigatruss (BORG): A mixed assembly architecture trade study
[15] She, Y., et al. (2020). In-orbit robotic assembly mission design and planning to construct a large space telescope
[16] Lee, N., et al. (2016). Architecture for in-space robotic assembly of a modular space telescope

Introduction: Optimization of Strut-Cable Structures

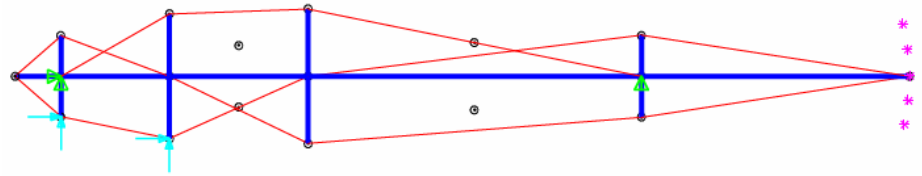
Tensegrity Topology Optimization [17]



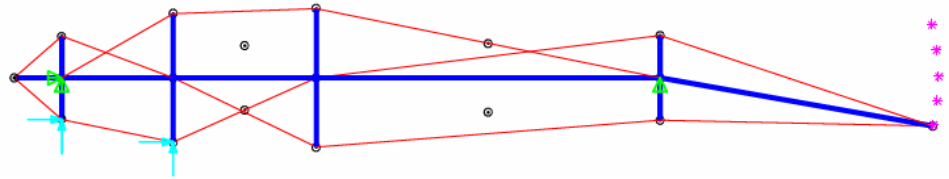
(a) Design space



(b) Ground structure

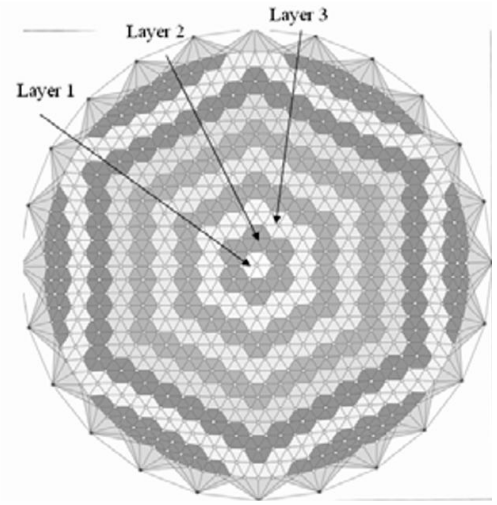


(c) Tensegrity at neutral trailing edge position

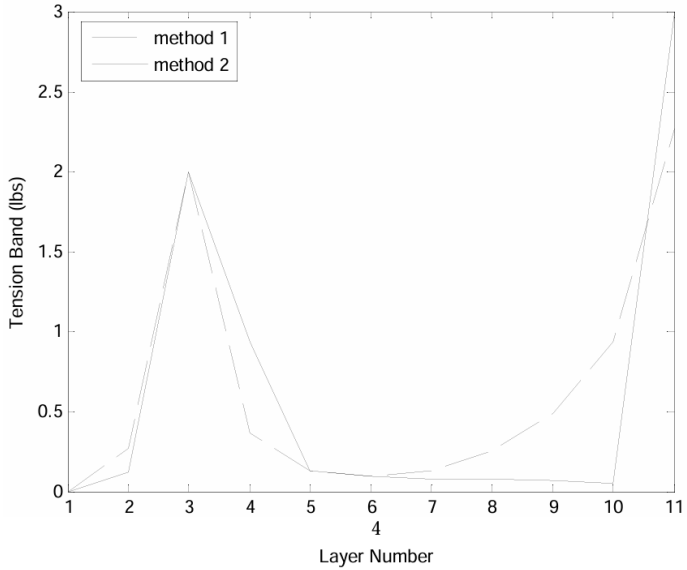


(d) Tensegrity at lowered trailing edge position

Cable Net Prestress Optimization [18]



Tension Band Distribution



[17] Myszka, D. H., et al. (2019). Topology optimization of cable-actuated, shape-changing, tensegrity systems for path generation
[18] Yang, B., et al. (2009). Optimal design of initial surface profile of deployable mesh reflectors via static modeling and quadratic programming

Prestressing Model: Geometry Definition [19]

User Inputs

- Reflector diameter D
- Focal length F (assumed to equal D)
- Allowable root mean square triangulation error δ

Number of Hexagonal Rings

- Spherical cap radius of curvature: $R = 2F + D^2/32F$
- Maximum equilateral triangle side length: $L_{max} = \sqrt{8\sqrt{15}\delta R}$
- Number of rings: $N = \text{ceil}(D/2L_{max})$, add 1 if odd
- Actual equilateral triangle side length: $L = D/2N$

Reflector Geometry

- Define inner net tessellation for triangular slice, rotate sixfold
- Define boundary net/longerons, flip about xy plane for rear net
- Connect nets vertically with battens and tension ties
- Calculate equilibrium matrix, ensure static and kinematic determinacy via SVD and Maxwell's rule

[19] Suh, J. E., et al. (2024). Scaling laws for deployable mesh reflector antennas

Prestressing Model: Tension Tie Force Optimization [19]

Basis Generation

- Equilibrium matrix of structure: A (rows = nodal DOFs, columns = structural elements)
- Construct vector t_i, with unit z-force for nodes on ring $i \in [1, N+1]$ (including central node of tessellation as a ring) and all other force components set to 0
- Solve $Ab_i = t_i$ for structural member forces for all i, create basis matrix $B = [b_1, b_2, \dots, b_{N+1}]$
- Split B into rows corresponding to cables, longerons, and battens: $B = \begin{bmatrix} B_c \\ B_l \\ B_b \end{bmatrix}$

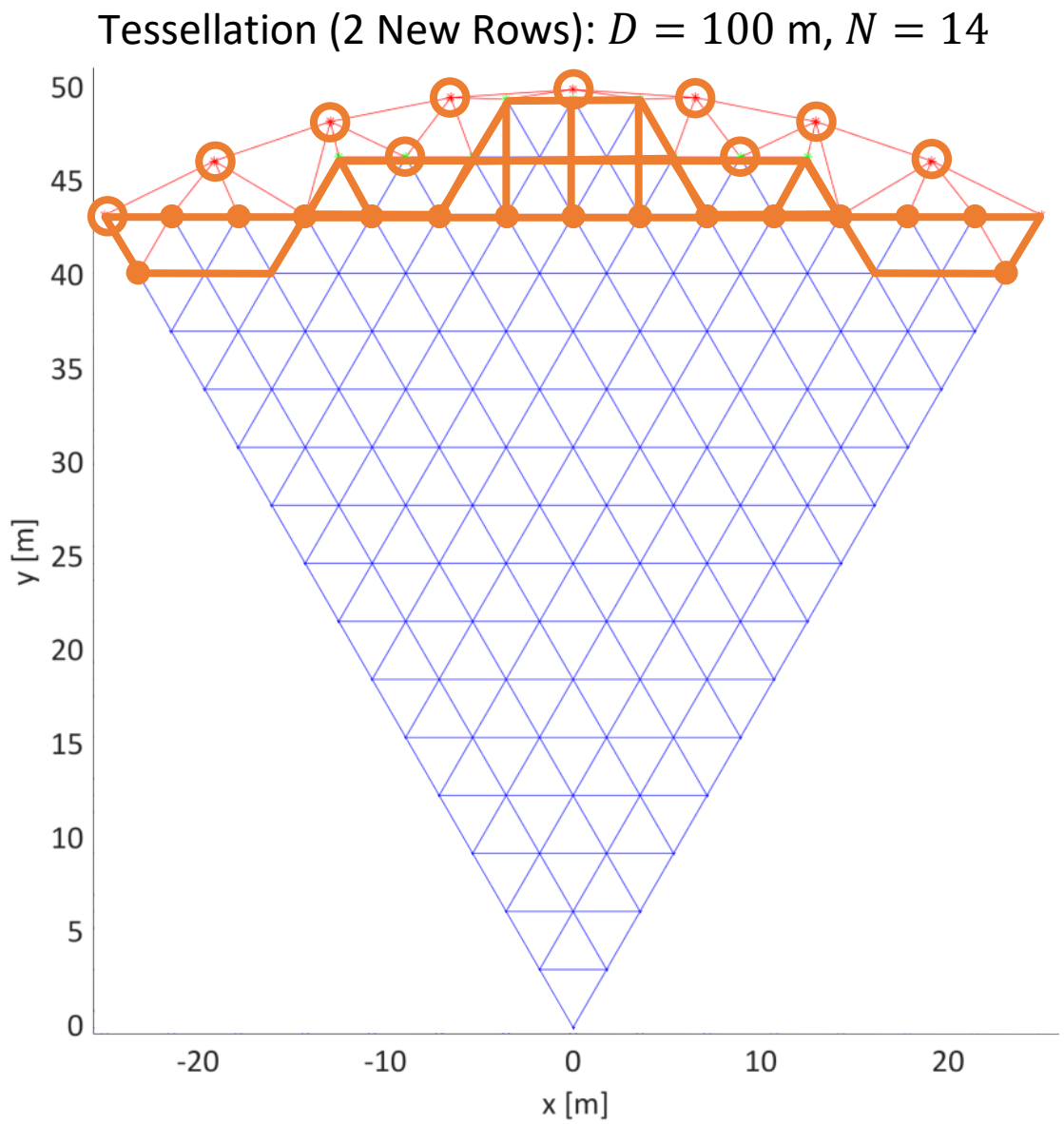
Prestress Optimization

- Goal: Find x (vector of tension tie forces in each ring) via linear programming (simplex method)
- Objective: Minimize average batten compressive force, $f(x) = \frac{-1}{n_b}\,[1, 1, \dots, 1]B_b x$, where n_b = number of battens in structure
- Constraints: $B_c x \geq c_{min}$ (all cables must exceed specified minimum tension to support mesh)
- After finding x, determine forces in all struts and cables as Bx

[19] Suh, J. E., et al. (2024). Scaling laws for deployable mesh reflector antennas

Prestressing Model: Adding Triangles

- "Step pyramid" method of adding triangles to fill in boundary area while maintaining distance from perimeter
  - Indent 1st row by 5 triangles to flip edge triangle orientation from "cup" to "pyramid"
  - Indent later rows by 4 triangles to maintain orientation
- Batten connection method preserved from original algorithm
  - Need to add even number of rows
  - 1 pair of added rows = 2 new inner-net edge nodes (requires 1 extra batten) compared to top row of original hexagonal tessellation
  - Per slice: $n_{b,slice} = \frac{N + n_r}{2}$



Prestressing Model: Optimization Behavior

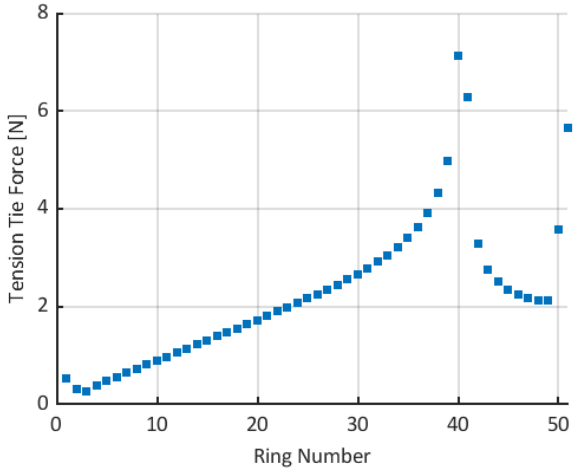
- Batten connection method requires even number of added rows
- For $N < 14$, 2nd added row goes beyond boundary of perimeter truss
- Tessellation adjustment enables 34% increase in upper diameter limit for convergence compared to original method
- 4+ added rows always produces infeasible optimization

D (for RMS = $\lambda/50$)	N	Optimization Behavior
$D < 22$ m	$N < 14$	Converges without added rows, cannot add rows
$D \in [22, 377]$ m	$N \in [14, 50]$	Converges with or without 2 added rows
$D \in [378, 507]$ m	$N \in [52, 58]$	Only converges with 2 added rows
$D > 507$ m	$N > 58$	No convergence with any number of added rows

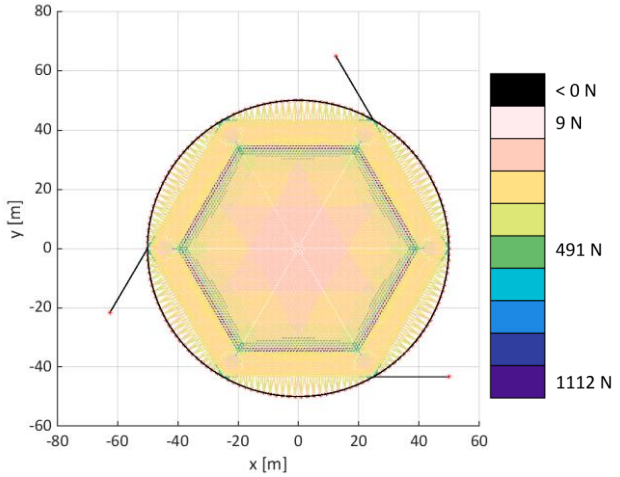
Prestressing Model: Limits for $D = 100$ m

$D = 100$ m, $N = 50$ (Limit Without Added Rows)

Tension Tie Forces

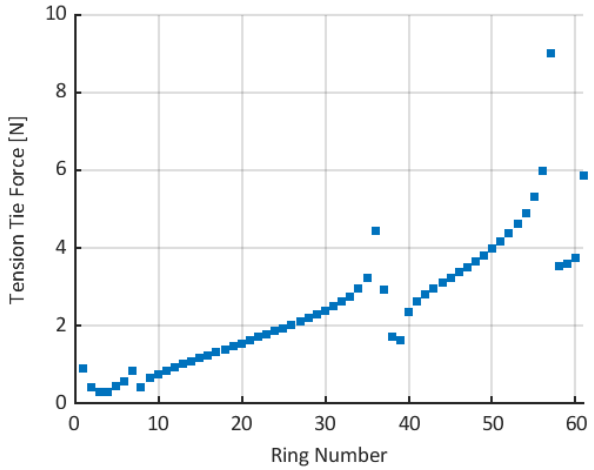


Cable Forces

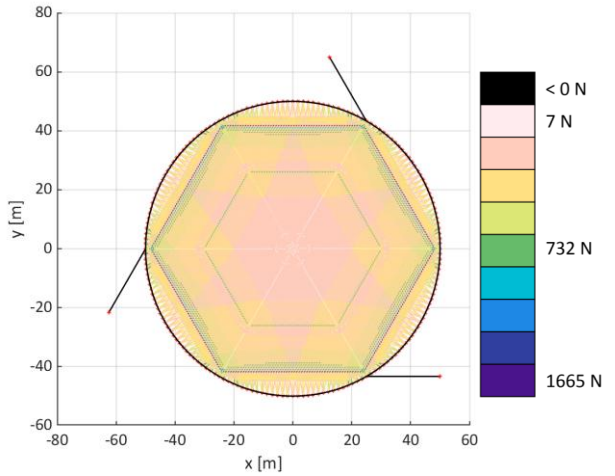


$D = 100$ m, $N = 58$ (Limit With 2 Added Rows)

Tension Tie Forces



Cable Forces

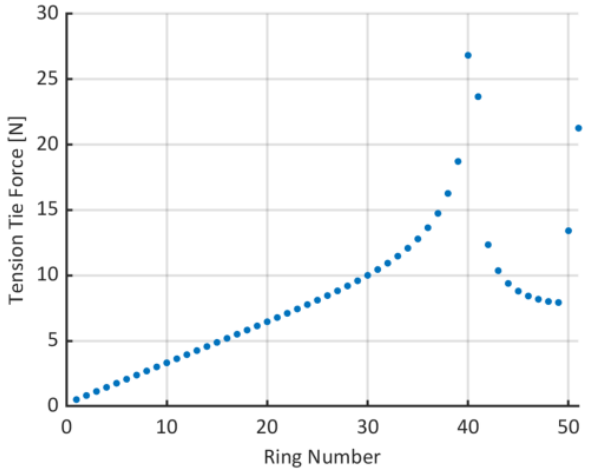


Caltech

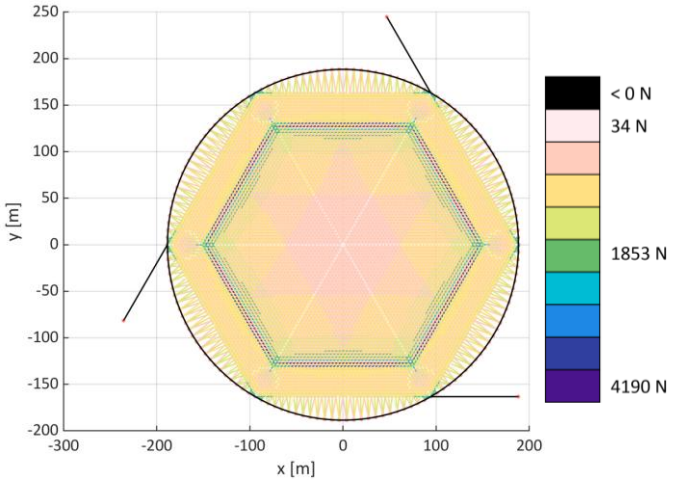
Prestressing Model: Limits for RMS Error = λ/50

$D = 377$ m, $N = 50$ (Limit Without Added Rows)

Tension Tie Forces

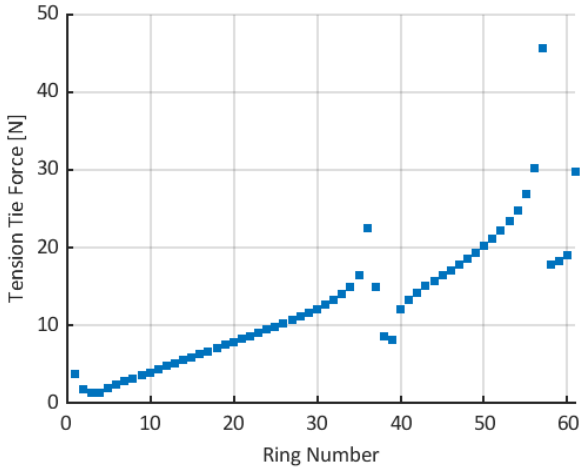


Cable Forces

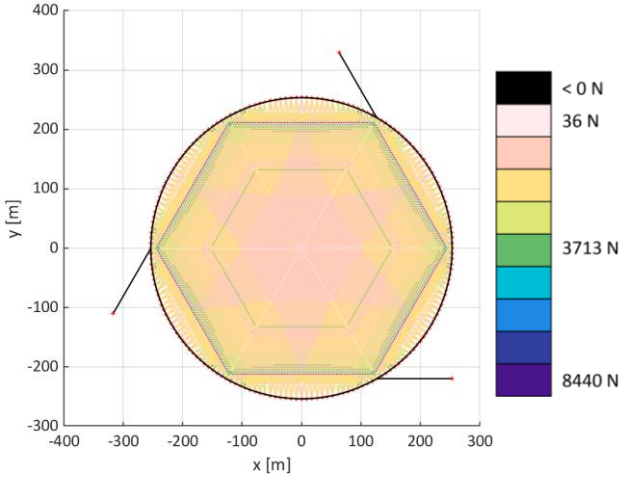


$D = 507$ m, $N = 58$ (Limit With 2 Added Rows)

Tension Tie Forces



Cable Forces

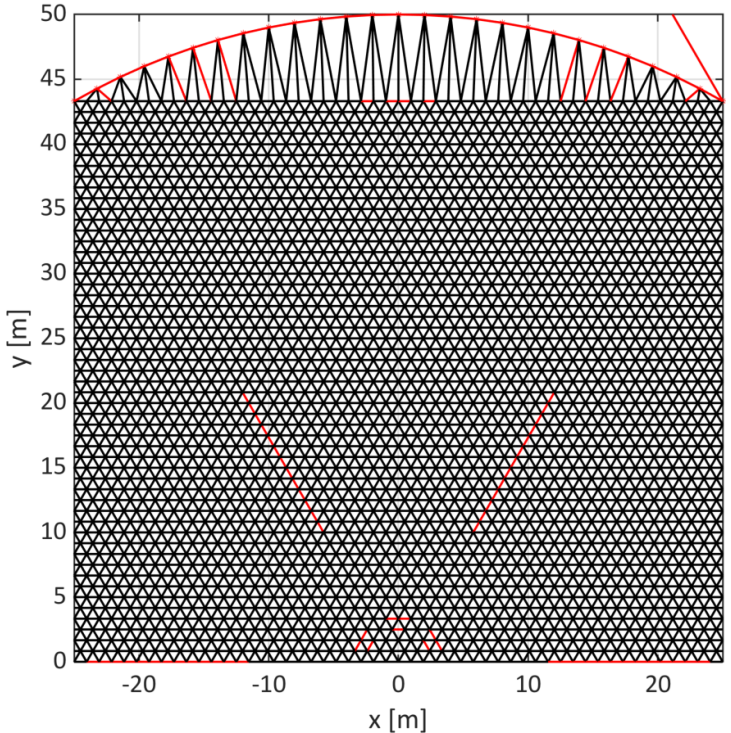


Caltech

Prestressing Model: Compressive Cables Beyond Limits

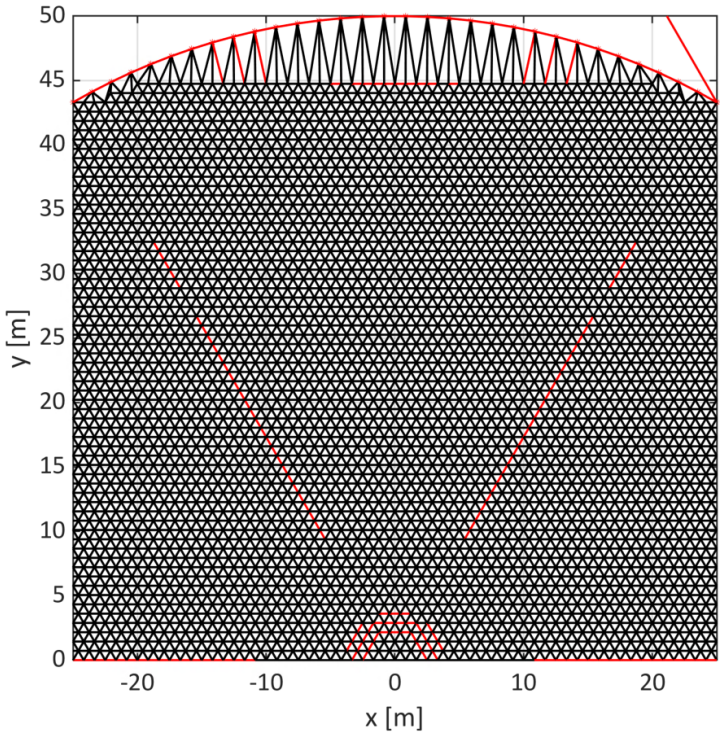
- Batten connection method requires even number of rings
- Used tension tie forces from limiting cases and duplicated innermost force 2x to provide a simple extension for 2 added rings
- Compressive cables emerge in same locations as stress concentrations in limiting cases, as well as near center due to force duplication
- Other extension methods from limiting cases provide similar results except at center

$D = 100$ m, $N = 52$





$D = 100$ m, $N = 60$ (2 Added Rows)

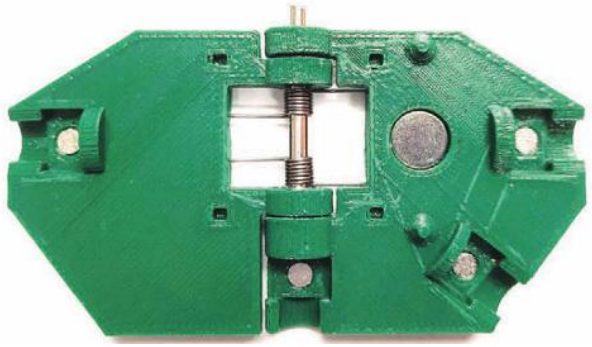


Caltech

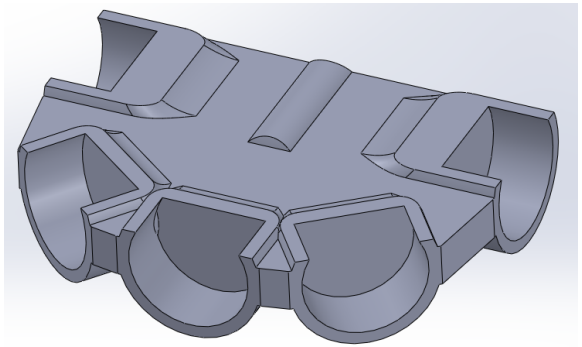
Prestressing Model: Joint Design for $D = 507$ m

- Joint mass found from CAD of simplified design with prescribed material and density (CFRP, 1780 kg/m^3)
- Used reference joint design for $D = 10$ m for 2 types of joints (strut diameter = 5 mm)

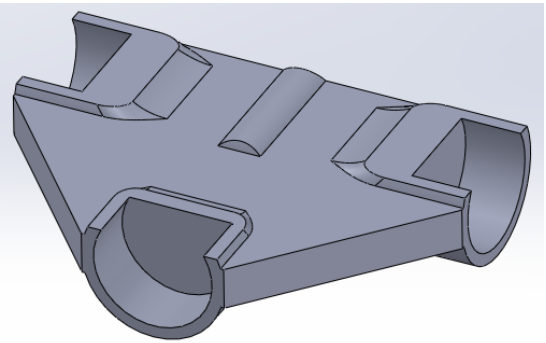
NOM4D Prototype Joint Design [20]



$D = 10$ m Type 1 Joint Design

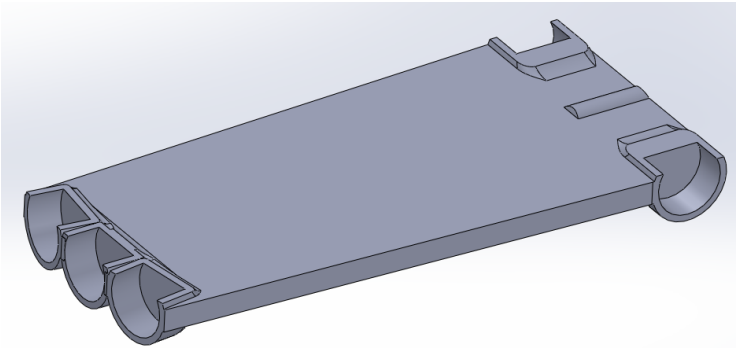


$D = 10$ m Type 2 Joint Design

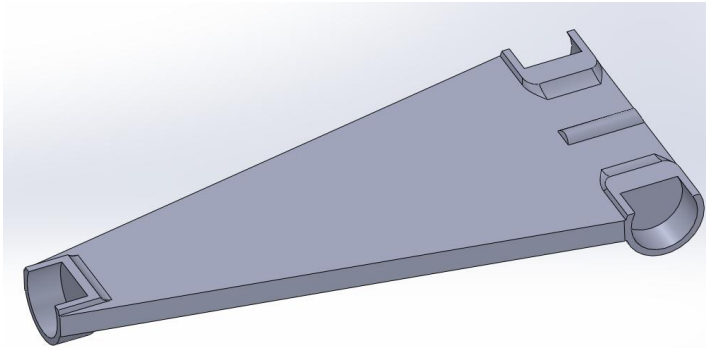


- Scaled dimensions for $D = 507$ m (strut diameter = 509.9 mm)

$D = 507$ m Type 1 Joint
Mass: 357.3 kg



$D = 507$ m Type 2 Joint
Mass: 269.4 kg



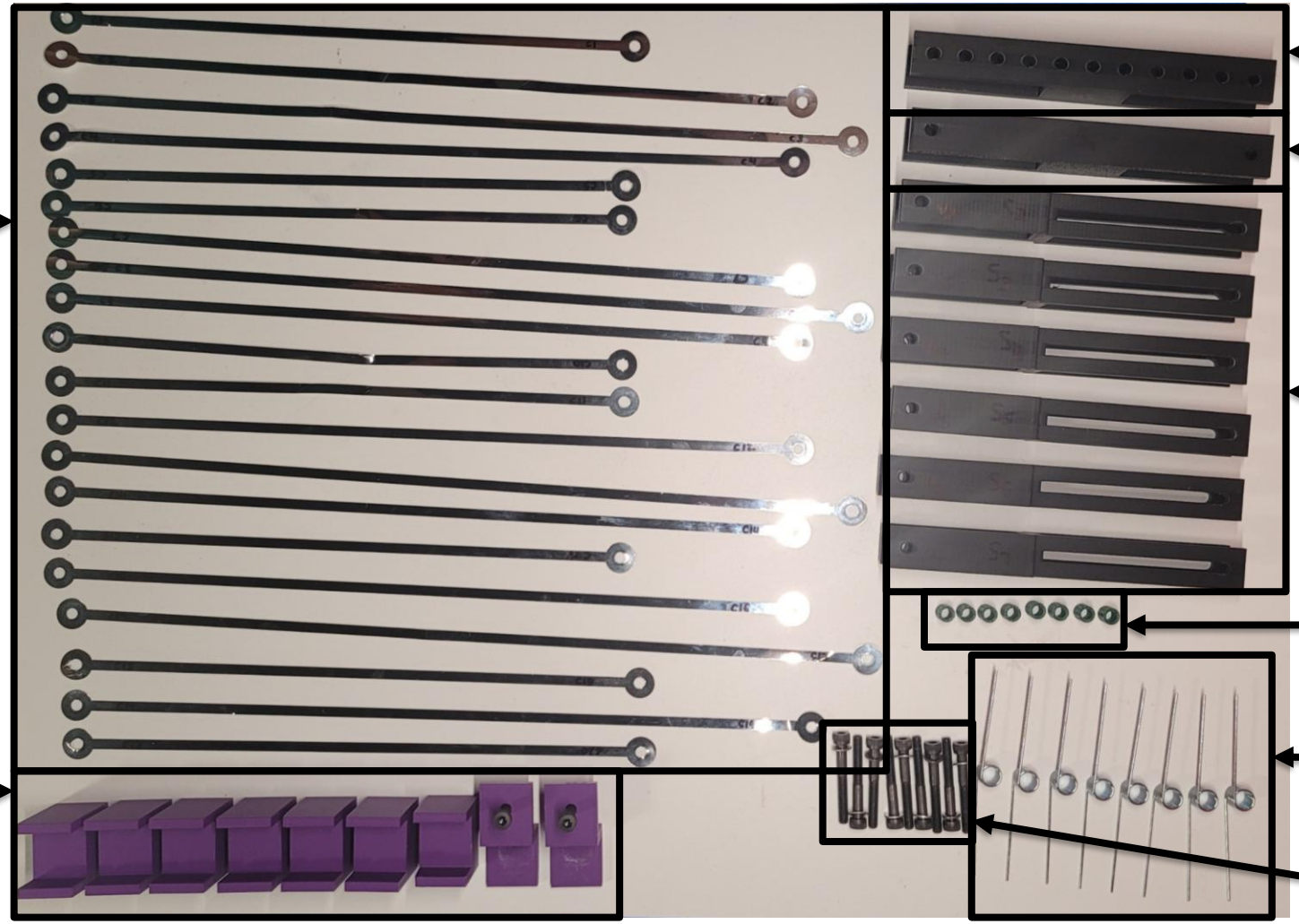
[20] Suh, J. E., et al. (2024). Concept for scalable mesh reflector antennas assembled in space

Prestressing Model: Mass for $D = 507$ m

- Material properties (stiffness, density) assumed for struts, mesh, cable net
- All struts have same radius to simplify joints (thickness varies to satisfy buckling criteria)
- Joints are 70% of total mass: Inefficient design for large reflectors, need to redesign

Component	Estimation Method	Mass [kg]	Percent of Total Mass [%]
Cable Net	Cable lengths with assumed cross-section area	522.0	0.3264
Mesh	Paraboloid area and material areal density	3075	1.923
Tension Ties	2x mesh mass	6150	3.846
Longerons	Specified thickness, radius from buckling criterion with 2x safety factor	4557	2.850
Battens	Same radius as longerons, thickness from buckling criterion with 2x safety factor	20430	12.78
Diagonals	No compression (present for rigidity): Same radius as longerons, 75% of longeron thickness	12380	7.740
Joints	Calculated from CAD	112,800	70.54
Total	**Sum of components**	**159,900**	**100**

Assembly Model: Prototype Components



Cables in 3 sizes to connect across 2-, 3-, and 4-vertex separations

Supports to raise model off table, for easier access to bolt holes during assembly

S_R with bolt holes for varying R_0

S_1 (constructed specially to interface with S_R)

Remaining struts

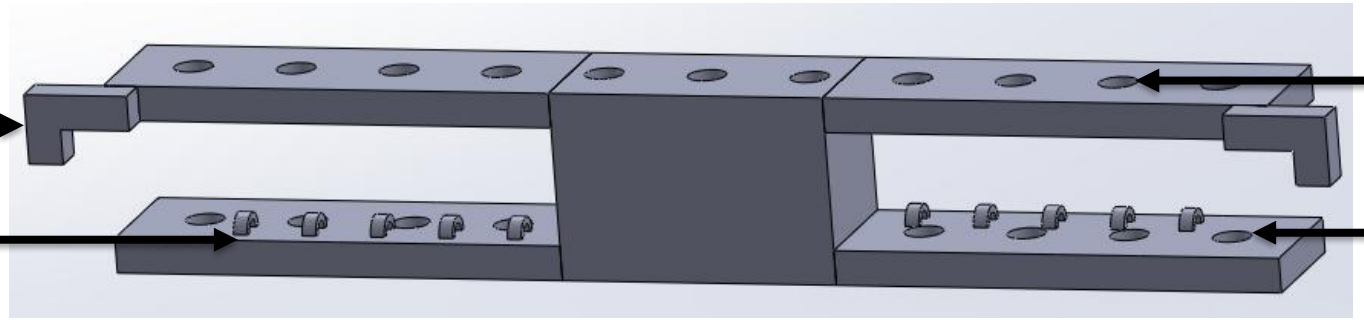
Collars inserted into torsional springs to ensure a tight fit with bolts

Torsional springs

Bolts with washers to secure torsional springs at joints

Assembly Model: Prototype Strut Design

S_R



Angle stop (optional)

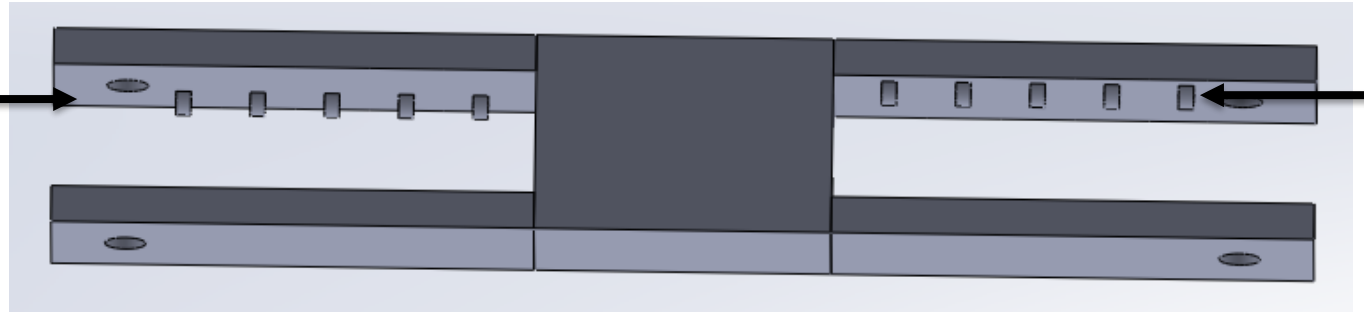
Loops to secure torsional spring end pieces in place

Bolt holes every $0.1L$ for varying R_0 placement

Bolt hole to secure torsional spring

S_1

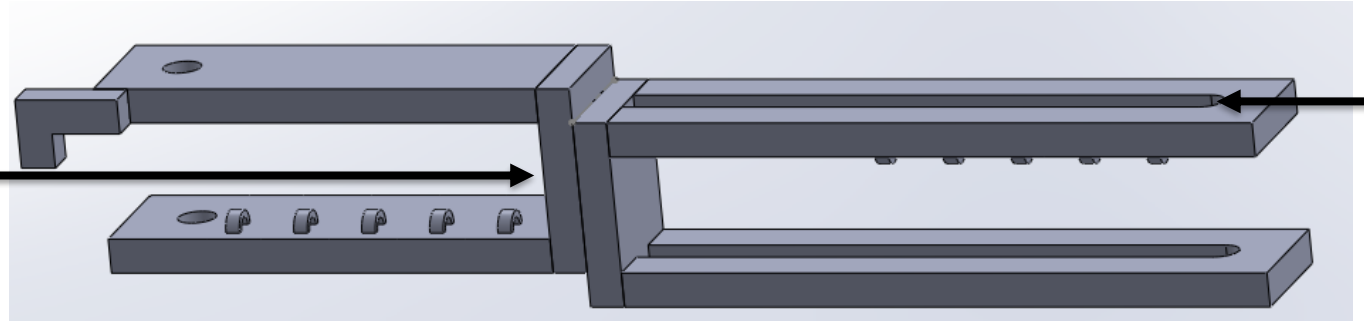
No need for angle stops due to alternating high-low strut positioning (S_1 is fully in low position, angle stops are on high-position struts)



Torsional spring loops on top arm of strut due to low positioning

Remaining Struts

Staggered high-low design to enable assembly of any number of struts (odd or even)

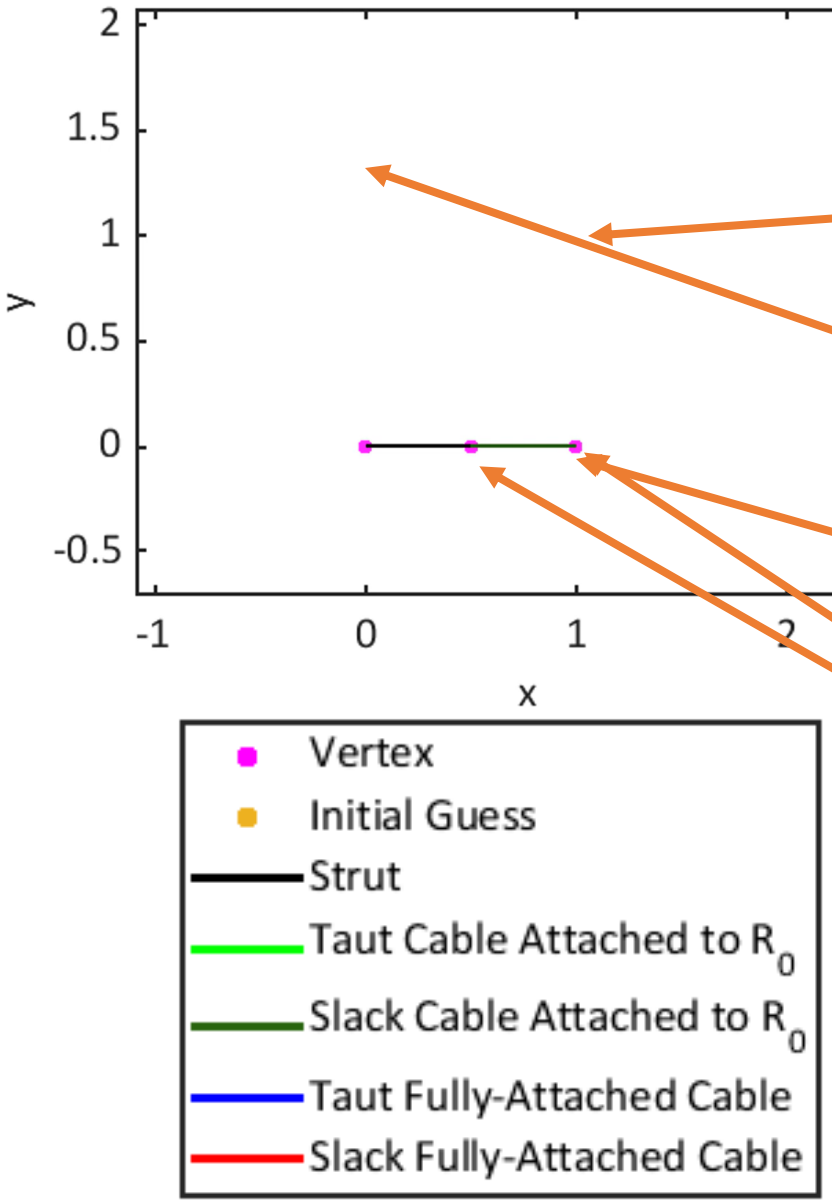


Slot to slide strut for testing of partial ejection from R_2

Caltech

Assembly Model: Octagon (No Angle Stops)

$R_0 = 0.5L$ MATLAB, No Convexity Constraint



Regular polygon due to slack cables

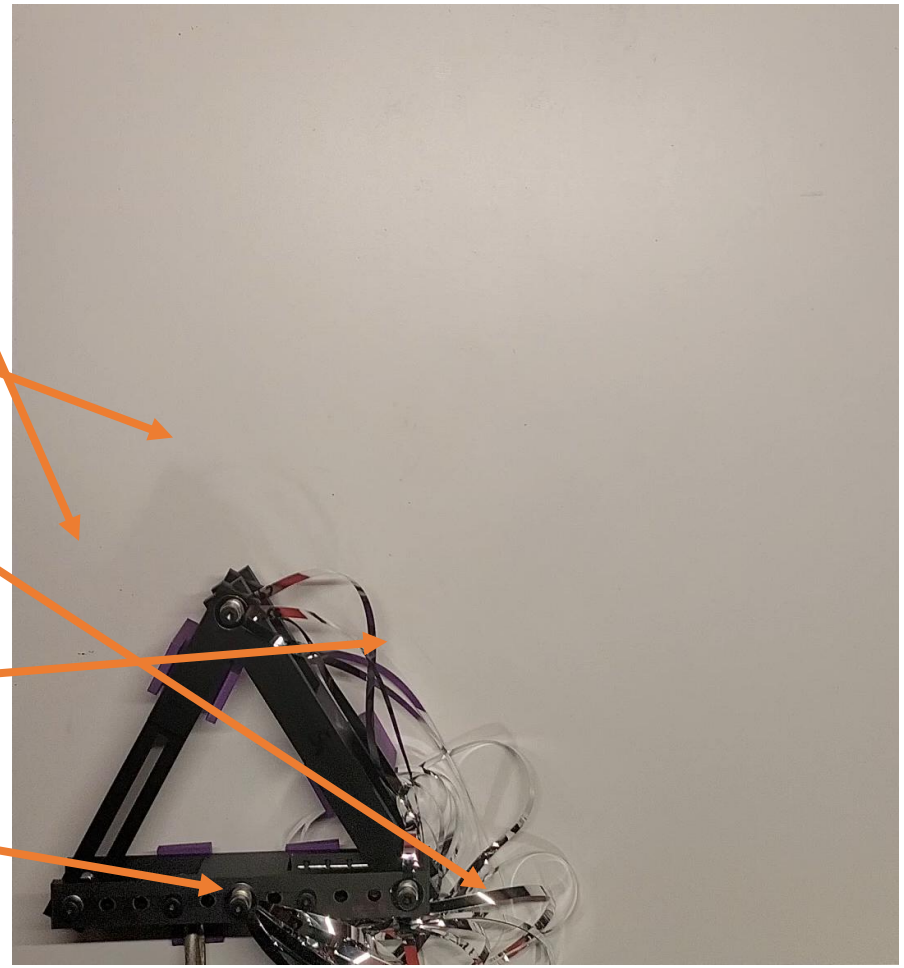
Irregular shape due to taut cables

Concave penultimate shape

Concave final shape

R_0

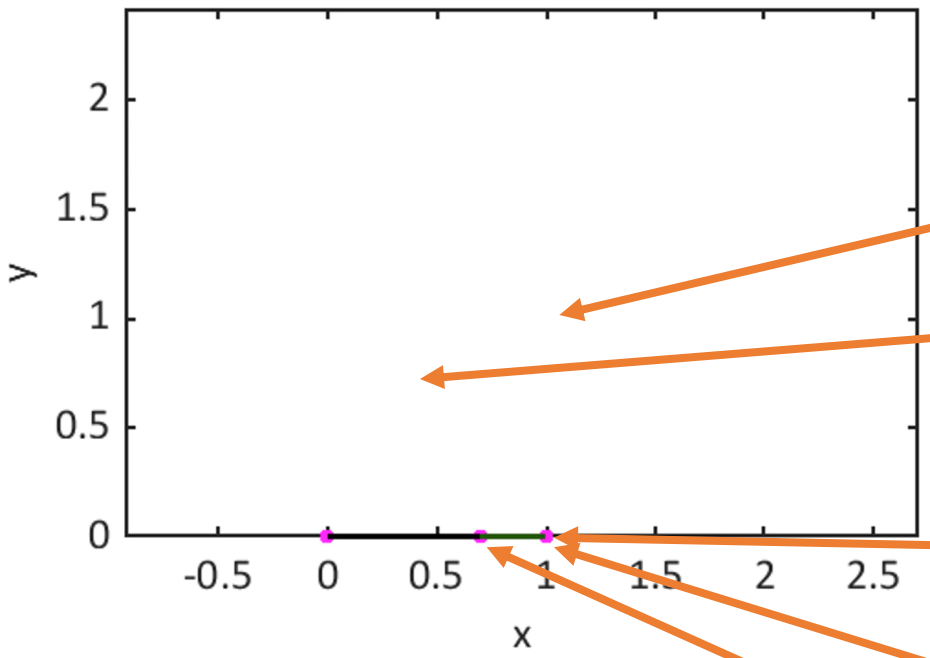
$R_0 = 0.5L$ Prototype, No Angle Stops

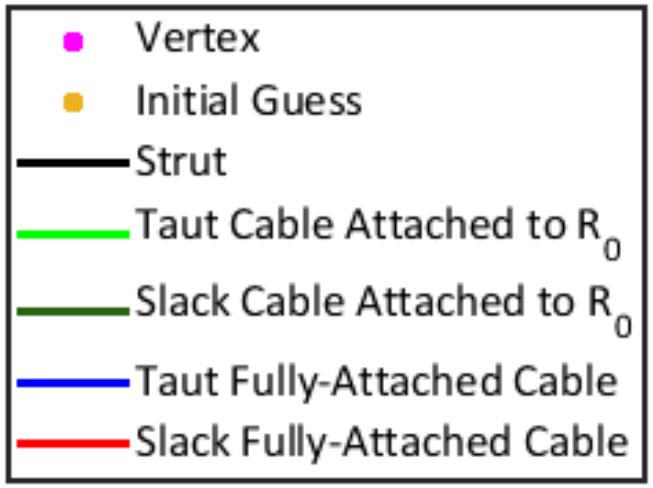


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Assembly Model: Octagon (Angle Stops)

$R_0 = 0.7L$ MATLAB, Convexity Constraint





Regular polygon due to slack cables

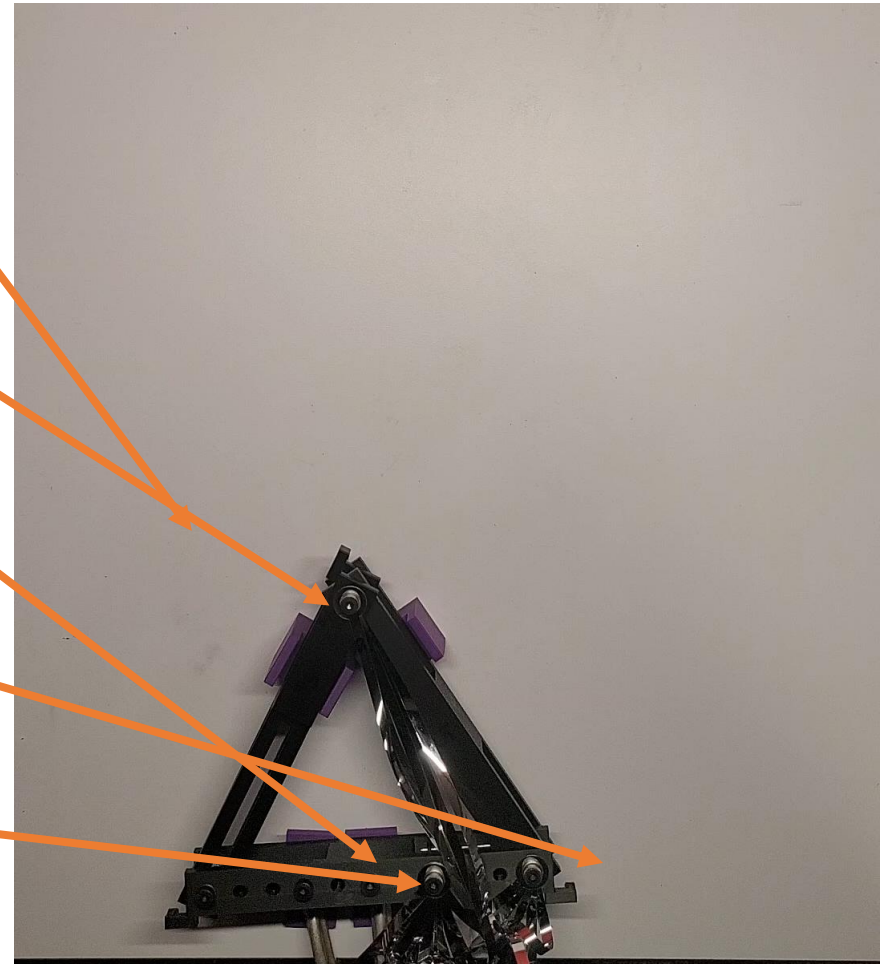
Irregular shape due to taut cables

Angle stops active

Desired regular octagon

R_0

$R_0 = 0.7L$ Prototype, Angle Stops



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Assembly Model: $R_0 = 0$ Feasibility Proof

Hypothesis

Setting $R_0 = R_2$ allows for a feasible sequence of regular polygons at all steps for any s, with the final step recovering the desired shape

Requirements for Feasibility at Each Step:

- All struts have length L (automatically satisfied by regular polygons)
- All angles satisfy convexity constraint (automatically satisfied by regular polygons)
- All cables have length less than or equal to taut length (need to prove)

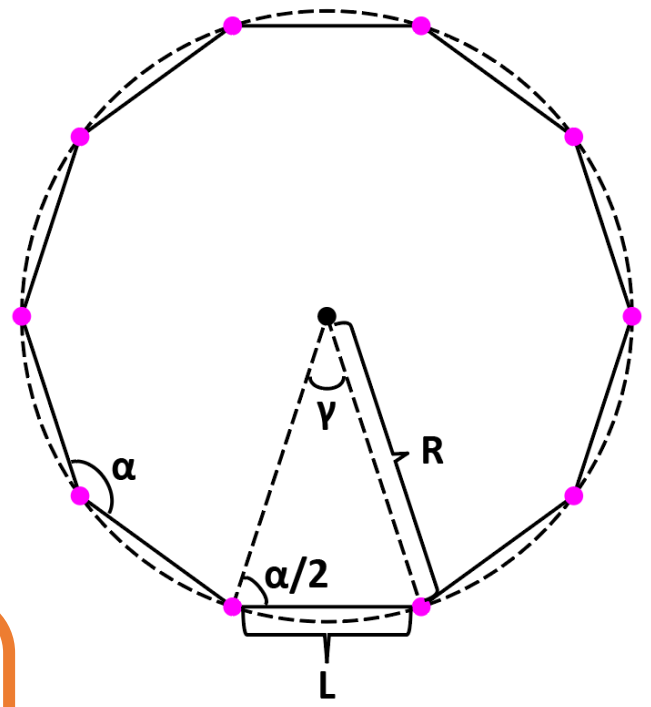
To Prove Cable Constraint Satisfaction:

- Calculate taut cable lengths from diagonal lengths of final regular polygon (so final shape automatically satisfies all constraints)
- Demonstrate that fully-attached cables at intermediate steps satisfy cable constraints
- Demonstrate that cables attached to R_0 at intermediate steps satisfy cable constraints

Assembly Model: $R_0 = 0$ Feasibility Proof

- Calculation of cable taut lengths:
  - Interior angles in final regular s-sided polygon: $\alpha = \pi(s-2)/s$
  - Central angle of circumscribed circle subtended by chord representing 1 side of polygon: $\gamma = 2\pi/s$
  - Law of sines for isosceles triangle connecting center of circle with 1 side of polygon: $\frac{\sin(\alpha/2)}{R} = \frac{\sin(\gamma)}{L}$
  - Solve for radius of circumscribed circle and simplify trigonometric identities: $R = \frac{L}{2\sin(\pi/s)}$
  - Central angle subtended by chord connecting across k-vertex separation: $\beta = 2\pi k/s$

Final Regular Polygon Schematic



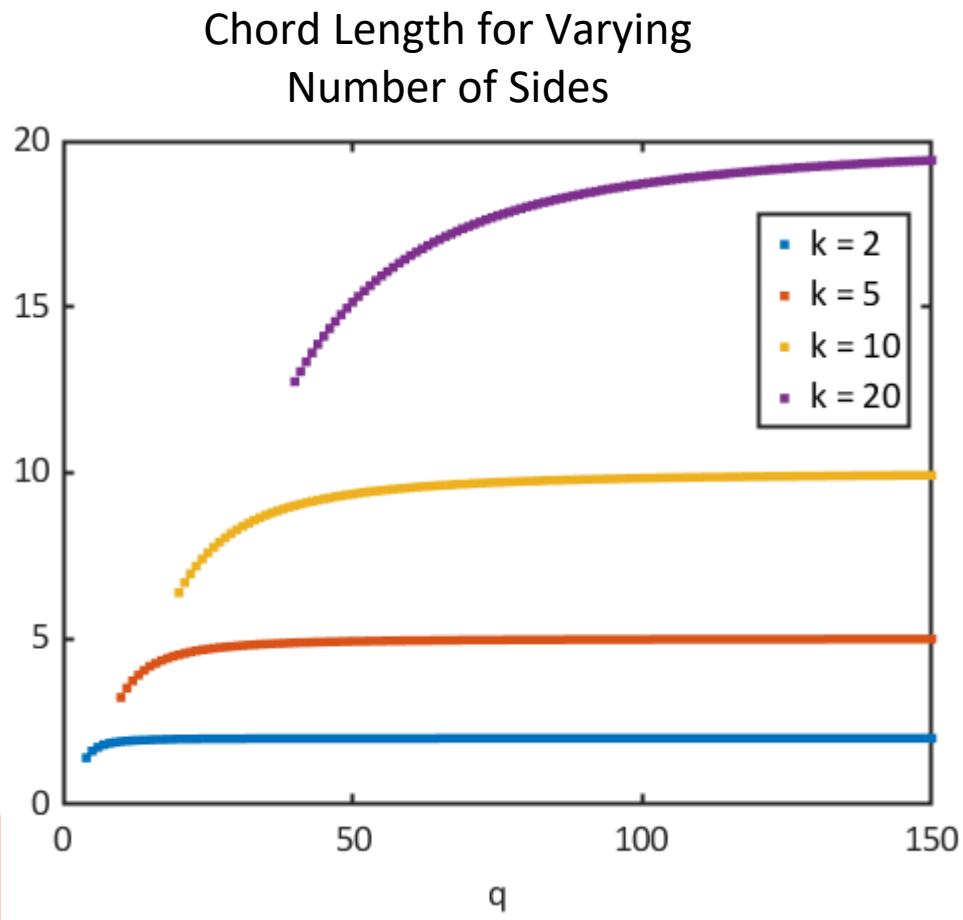
Taut length of cable with k-vertex separation in an s-sided polygon with side length L (based on known equation for chord length of a circle):

$$c(s,k,L) = 2R\sin(\beta/2) = \frac{L\sin(\pi k/s)}{\sin(\pi/s)}$$

Assembly Model: $R_0 = 0$ Feasibility Proof

- Fully-attached cable constraint satisfaction at intermediate steps:
  - Length of fully-attached cable across k-vertex separation in q-sided intermediate polygon: $c(q;k,L) = \frac{L\sin(\pi k/q)}{\sin(\pi/q)}$
  - A q-sided intermediate polygon has chords with vertex separations $k \in \{2,3,\ldots,\text{floor}(q/2)\}$, so chords with a given k-vertex separation only exist for $q \geq 2k$
  - Plotting $c(q;k,L)$ vs. q for various k and fixed L demonstrates that c increases monotonically with q (horizontal asymptote at $c = Lk$)

A fully-attached cable in an intermediate q-sided polygon will have a length less than or equal to its taut length in the final s-sided polygon for $q \leq s$



Assembly Model: $R_0 = 0$ Feasibility Proof

- R_0-attached cable constraint satisfaction at intermediate steps:
  - Cable $C_{r,t}$ connects V_r to V_t, where $t - r > 1$
  - $C_{r,t}$ is attached to $R_0 = R_2 = V_q$ at a step forming a q-sided polygon if V_r has been ejected and V_t has not, i.e. $r < q \leq t$
  - Need to show that vertex separation between V_r and V_q is less than or equal to final k-vertex separation in same direction as separation between V_r and V_t

Counterclockwise:

- Counterclockwise vertex separation from V_r to V_t is $t - r = k$
- Counterclockwise vertex separation from V_r to V_q is $q - r$
- Subtract r from all terms of $r < q \leq t$: $0 < q - r \leq t - r$
- Vertex separation $\leq k$

Clockwise:

- Clockwise vertex separation from V_r to V_t is $(r + s) - t = k \rightarrow k - r = s - t$
- Clockwise vertex separation from V_r to V_q is r
- $r \leq k$ because $t \leq s$ (final vertex in final polygon is V_s)
- Vertex separation $\leq k$

Assembly Model: $R_0 = 0$ Feasibility Proof

- R_0-attached cable constraint satisfaction at intermediate steps:
  - Need to show that cable length increases monotonically with vertex separation, so if separation $< k$ due to R_0 attachment, cable has more slack than if it were fully connected
    - Cable length for $(k-1)$-vertex separation: $c(q, k-1, L) = c(q, k, L)\cos(\pi/q) - \cos(\pi k/q)$
    - For integer $q \geq 3$ and $k \in \{2,3, \ldots, \text{floor}(q/2)\}$, $\cos(\pi/q) \in (0,1)$ and $\cos(\pi k/q) \in [0,1)$
    - Thus, $c(q, k-1, L) < c(q, k, L)$ (same logic holds recursively for $k-2$, $k-3$, etc.)

A cable attached to R_0 in an intermediate step will have a length less than or equal to its taut length in the final polygon due to: (1) vertex separation being less than or equal to final k-vertex separation, and (2) smaller vertex separations corresponding to cables with more slack

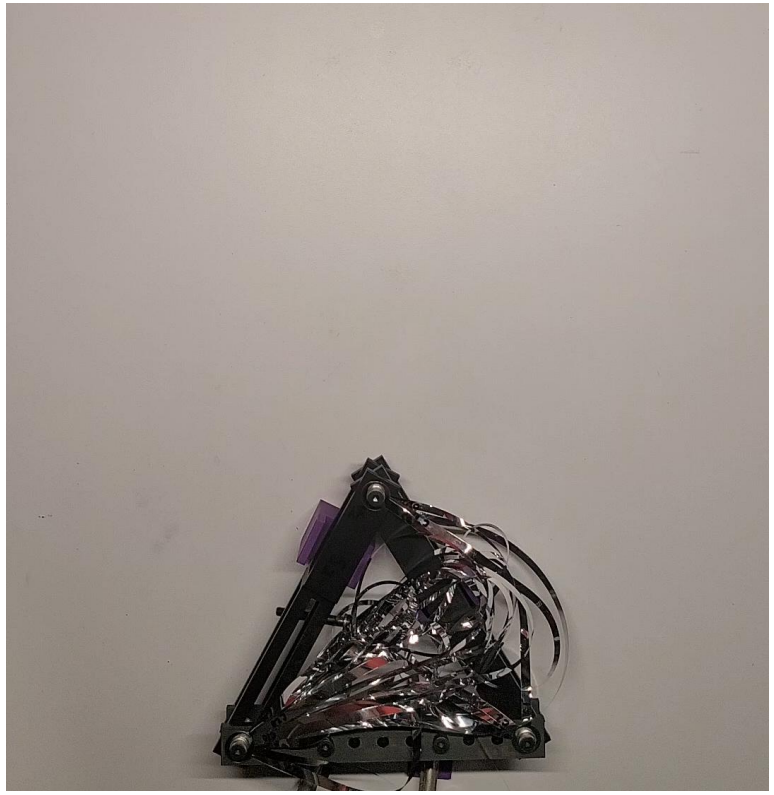
Conclusion

For an s-sided regular polygon with cables on all diagonals and $R_0 = R_2$, the set of q-sided regular polygons ($q \leq s$) satisfies all constraints at each step and therefore guarantees the existence of a successful assembly sequence for any s

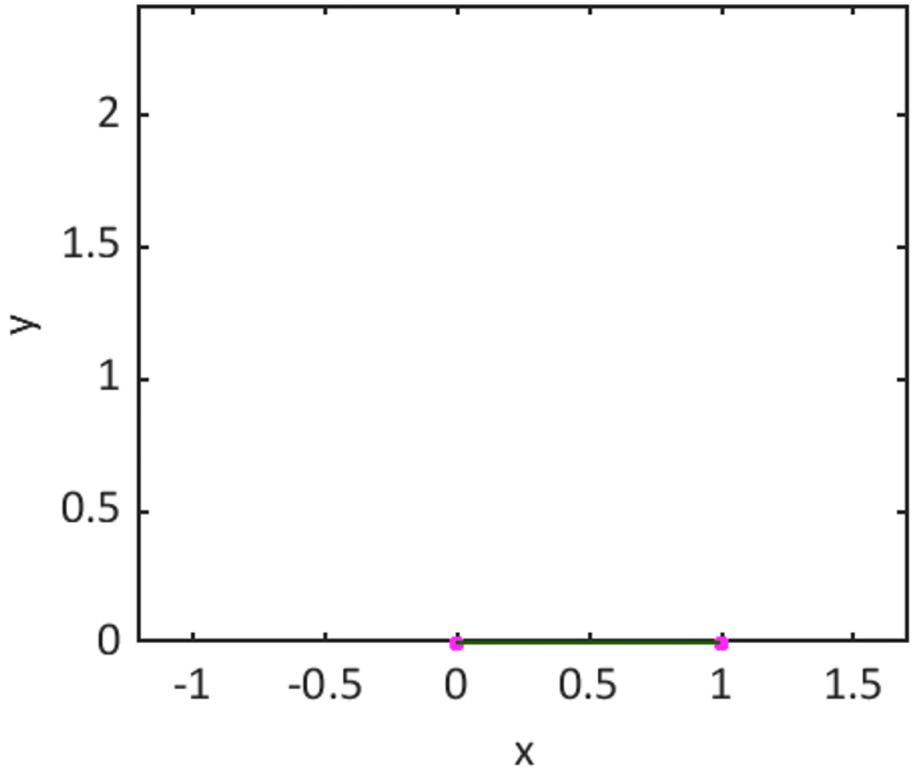
Assembly Model: $R_0 = 0$ Octagon Videos

- Regular polygons at each step, as expected
- Prototype requires repeated attachment/detachment of cables from $R_0 = R_2$
  - Demonstrates practical difficulties associated with having parking spot coincide with strut/cable attachment and ejection location

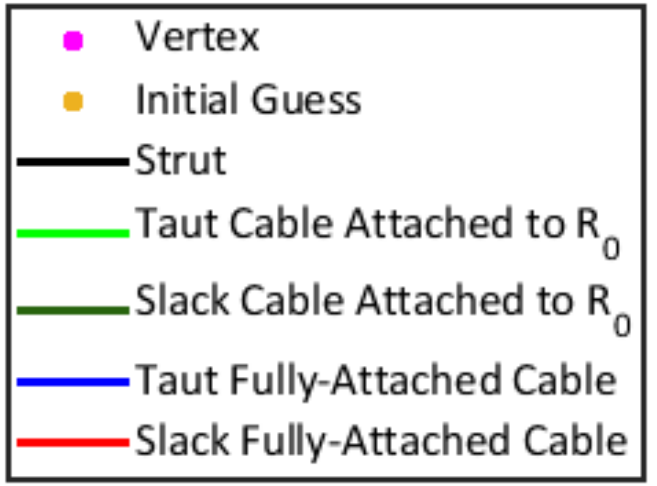
Prototype



MATLAB Model



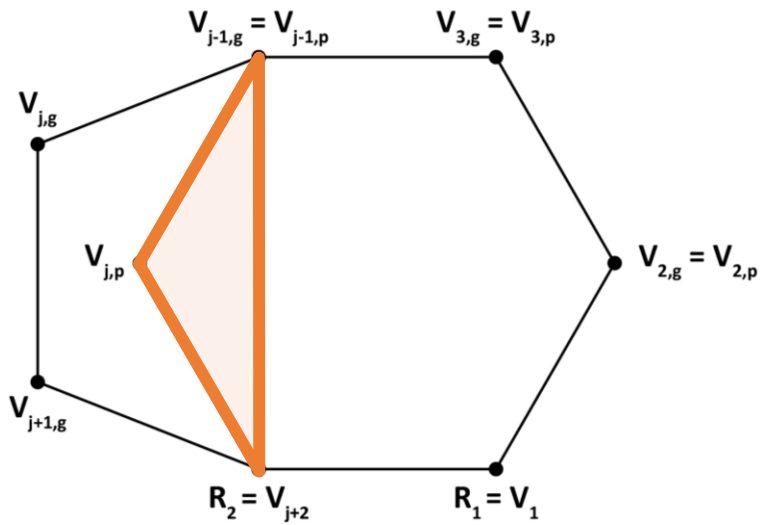
MATLAB Model Legend



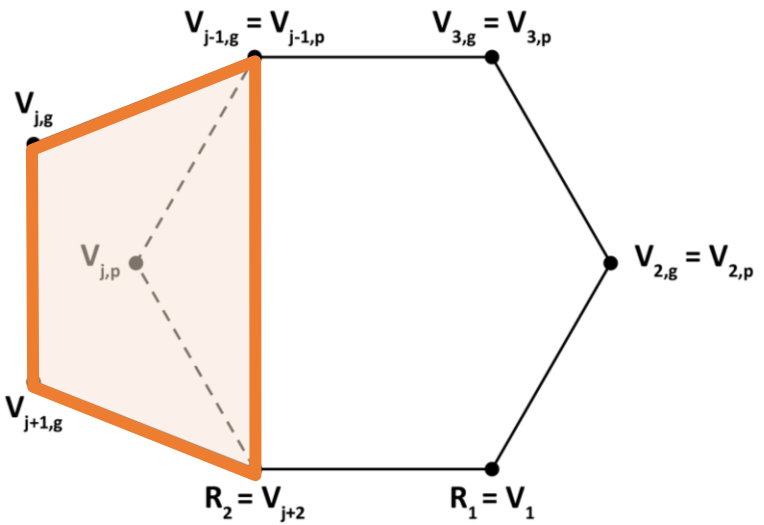
Assembly Model: Guess Overview

- 1st step guess = hardcoded, last step guess = penultimate shape
- For Step j, take guesses for V_2 to V_{j-1} as positions in converged solution from Step $j-1$
- Convert triangle between $V_{j-1,p}$, R_2, and $V_{j,p}$ into a trapezoid between $V_{j-1,g}$, R_2, and guesses of V_j and V_{j+1}
- Trapezoid should have one base and both legs of length L to satisfy strut length constraints (brings guess close to feasible region of optimization problem)
- Cable length constraints may not be satisfied by guess – optimization is responsible for locating a feasible solution near guess if one exists

Shape from Step $j-1$



Guess for Step j

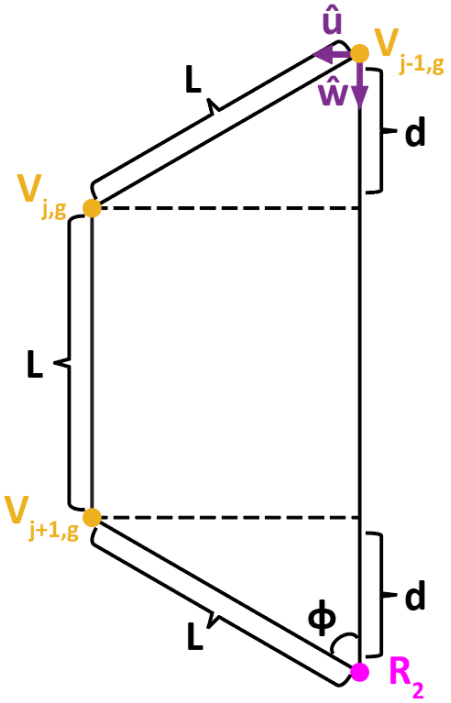


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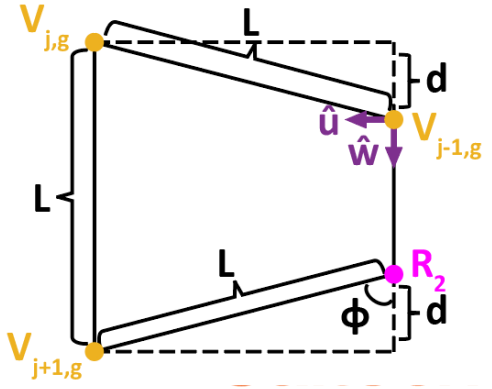
Assembly Model: Guess Details

- Positions of R_2 and $V_{j-1,g}$ are known from prior step
- Need to find $V_{j,g}$ and $V_{j+1,g}$ based on distances parallel/perpendicular to line connecting R_2 to $V_{j-1,g}$
- Define geometric parameters:
  - Length $d = \left|0.5\left(\left|R_2 - V_{j-1,g}\right| - L\right)\right|$
  - Angle $\phi = \arccos(d/L)$
  - Orthogonal unit vectors $\hat{w} = \frac{R_2 - V_{j-1,g}}{\left|R_2 - V_{j-1,g}\right|}$ and $\hat{u} = \left((\hat{w})_y, -(\hat{w})_x\right)$
- If $\left|R_2 - V_{j-1,g}\right| > L$:
  - $V_{j,g} = V_{j-1,g} + d\hat{w} + L\sin(\phi)\hat{u}$
  - $V_{j+1,g} = V_{j-1,g} + (L + d)\hat{w} + L\sin(\phi)\hat{u}$
- If $\left|R_2 - V_{j-1,g}\right| \leq L$:
  - $V_{j,g} = V_{j-1,g} - d\hat{w} + L\sin(\phi)\hat{u}$
  - $V_{j+1,g} = V_{j-1,g} + (L - d)\hat{w} + L\sin(\phi)\hat{u}$

$\left|R_2 - V_{j-1}\right| > L$



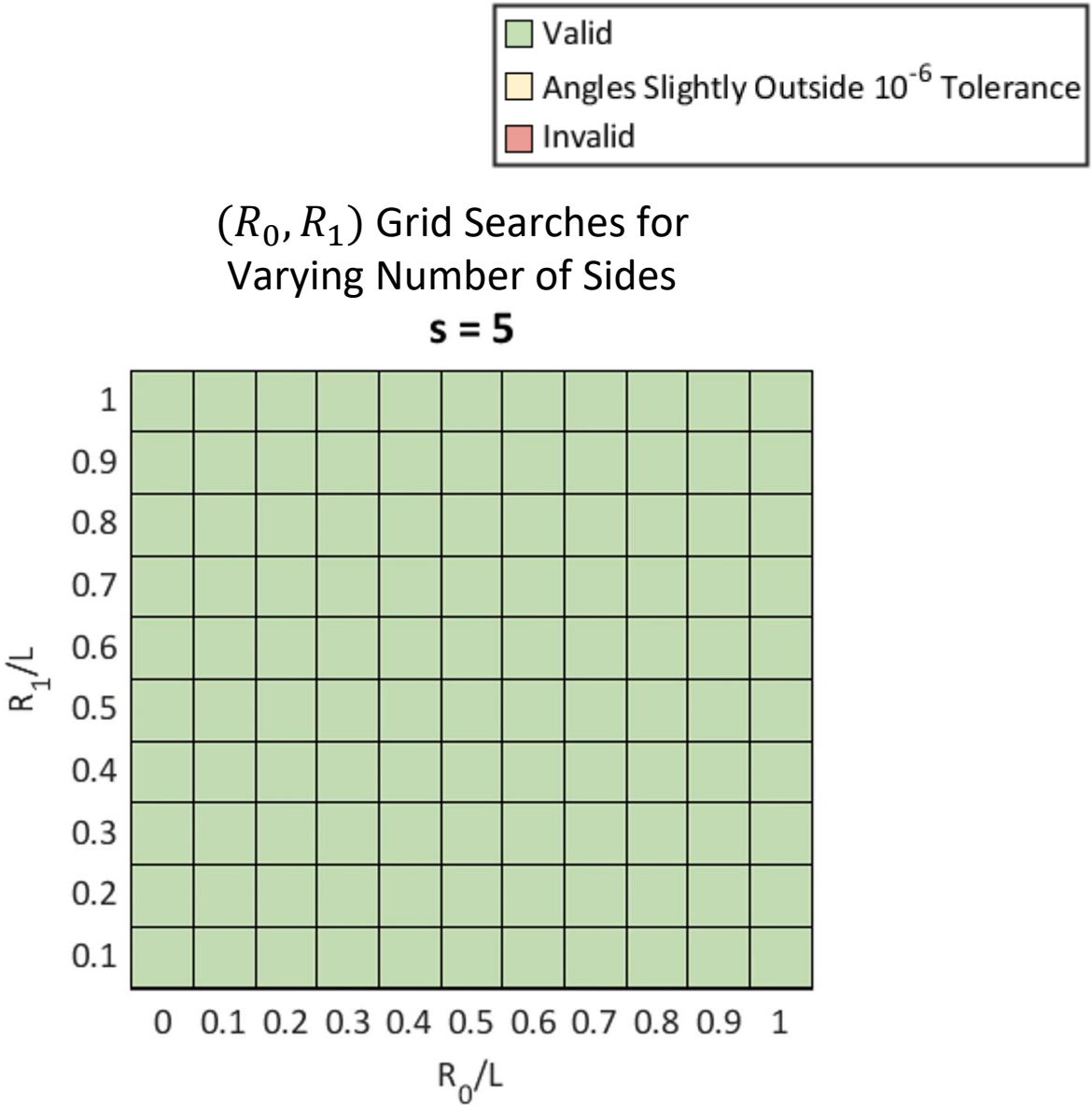
$\left|R_2 - V_{j-1}\right| \leq L$



Assembly Model: (R_0, R_1) Grid Searches

- Clear "staircase" structure of feasible cases
- As s increases, number of valid cases (and corresponding maximum R_0) decreases
  - Any $R_0 > 0.5L$: $s = 8$
  - $R_0 = 0.5L$: $s = 9$
  - $R_0 = 0.4L$: $s = 10$
  - $R_0 = 0.3L$: $s = 12$
  - $R_0 = 0.2L$: $s = 15$
  - $R_0 = 0.1L$: $s = 23$
- For all $s > 23$, only $R_0 = 0$ is feasible
- Grid searches are discrete – if mesh is refined further, may be able to assemble larger s for some $R_0 > 0$

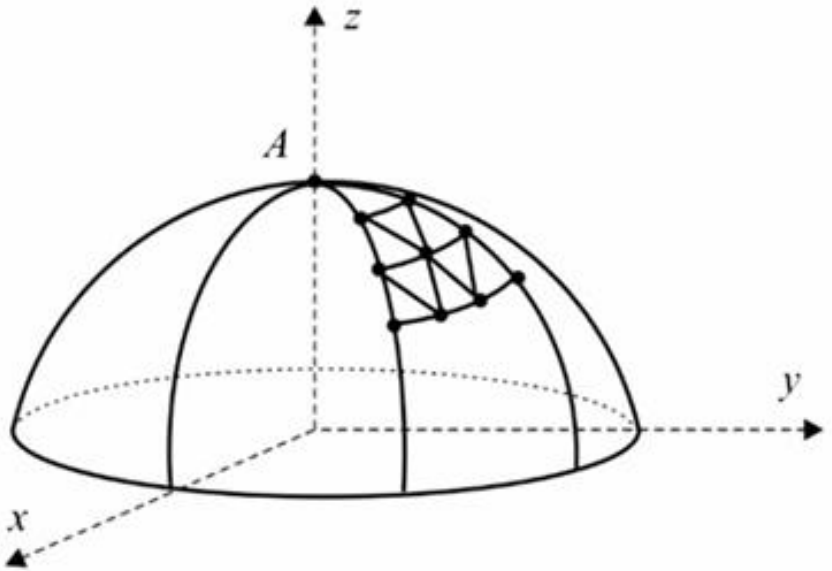
(R_0, R_1) Grid Searches for Varying Number of Sides



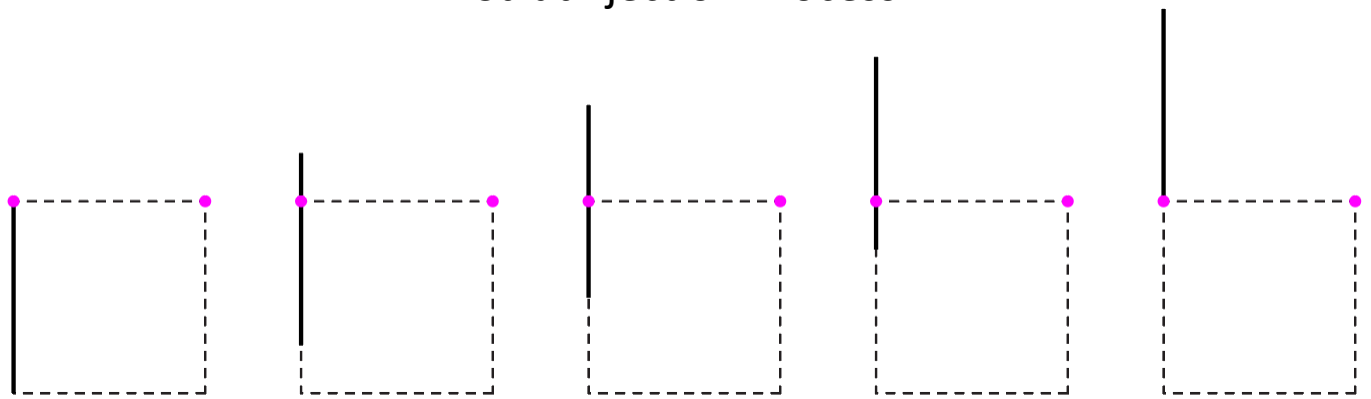
Future Work: Details of Next Steps

- Alternative tessellation methods for prestressing:
  - Latitude-longitude tessellation on spherical cap, projected onto paraboloid [21]
  - DistMesh software [22]: Input any tessellation as initial mesh, iterate to produce more equilateral triangles
- Improvements to physical realism of assembly model:
  - Continuous strut ejection rather than discrete
  - Incorporate elasticity in cables to correspond to Mylar behavior and reduce MATLAB/prototype discrepancies

Latitude-Longitude Tessellation on a Spherical Cap [21]



Strut Ejection Process



[21] Shi, H., et al. (2018). New methodology of surface mesh geometry design for deployable mesh reflectors
[22] Persson, P. O., & Strang, G. (2004). A simple mesh generator in MATLAB

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